

Broadcom

Outperform

Initiation: All stars aligned; best ASIC growth

Key Points

- ASIC is growing faster than GPU, driven by adoption among hyperscalers and vertical AI markets.
- Broadcom holds a near-monopoly in AI and cloud networking ASIC, with rising AI model complexity to boost its silicon content per rack.
- We initiate on Broadcom with an Outperform rating and an implied 17% total shareholder return to our US\$420 target price.

- **ASIC outpacing GPUs, broadening beyond hyperscalers.** Broadcom's ASIC technology is growing faster than GPU. We expect surging demand for ASIC to outpace GPU growth, driven by adoption among hyperscalers and vertical AI markets (eg, auto, healthcare, financial services). We estimate a 72% global AI ASIC market CAGR over 2025-28, with Broadcom capturing >70% market share.
- **Networking dominance.** Broadcom holds a near-monopoly in AI ASIC and cloud networking solutions, with leading positions in Ethernet switching, routing, and SerDes IP. Increasing AI model complexity is driving 3-5x growth in system-level bandwidth per generation, significantly boosting its silicon content per rack.
- **High-margin software expansion.** After its VMware acquisition, OPM rose from 62% (FY23) to 66% (3Q25). Its software business enhances long-term margin stability, high-quality free cash flow, and re-rate.
- **Valuation is justified by quality growth.** We believe Broadcom should trade at a premium to industry peers (Fig 60) with 1) strong growth outlook, 2) a strong ~34% dividend CAGR in recent years, and 3) long-term strategic planning supported by a unique management incentive plan.

Valuation: Our US\$420 TP is based on 33.5x 2027E EPS, based on a blended PE incorporating average multiples from semi solutions and software peers. We forecast 2025E/26E/27E earnings growth of 39%/41%/32%.

Catalysts: Better-than-expected AI capex plan from CSPs.

Investment Thesis and Recommendation

Broadcom dominates the rapid-growth AI ASIC market and SerDes/IP moat secures long-term edge in custom silicon, while its software business adds margin resilience and recurring visibility. We also like its disciplined, value-accretive M&A track record, yet it trades below sector averages.

Semicons & Semicon Equip United States



Arthur
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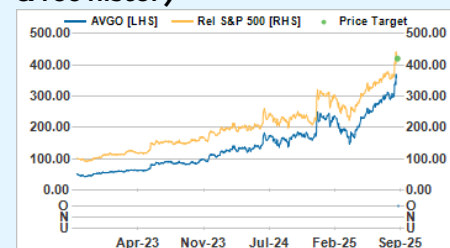
Paul
Golding

AVGO US	Outperform
Price (at 12 Sep 2025)	USD359.87
12-month target	USD420.00
12-month TSR (%)	17.3
Market Cap (Local) (m)	1,699,438
Market Cap (USD) (m)	1,699,438
Free Float (%)	98
30-day avg turnover (USD) (m)	8,778.0

Investment Fundamentals

Year end 31 Oct	2024A	2025E	2026E	2027E
Revenue (bn)	51.6	63.5	88.2	113.9
EBITDA (bn)	40.7	50.2	65.2	81.6
EBITDA growth (%)	57.0	23.3	29.8	25.2
EBIT (bn)	30.7	41.6	57.8	75.6
EBIT growth (%)	38.9	35.4	39.0	30.7
Reported profit (bn)	23.7	33.5	47.3	62.4
EPS adj [\$]	2.9	6.8	9.5	12.6
EPS adj growth (%)	582.7	134.7	40.6	31.9
PER adj (x)	124.8	53.2	37.8	28.7
Total DPS [\$]	1.1	2.3	2.3	2.3
Total DPS growth (%)	0.5	0.1	0.0	0.0
Total div yield (%)	0.3	0.6	0.6	0.6
ROA (%)	25.8	24.5	29.8	31.4
ROE (%)	51.8	45.4	48.3	44.1
EV/EBITDA (x)	44.6	36.7	28.3	22.6
Quant (rank vs. global sector)	2 / 664			

AVGO US rel S&P 500 performance, & rec history



Source: FactSet, Macquarie Research, Sep 2025 (all figures in USD unless noted, TP in USD)

Key Risks to Investment Thesis

- Hyperscaler AI capex normalisation may slow growth, but order books and tech roadmap support share dominance through 2027.
- Heavy reliance on hyperscalers; losing a major design win or insourcing could hit growth and margins.
- Export controls, shocks, or weak China demand may disrupt revenue.

Company Description

Broadcom supplies mixed-signal semiconductor components to smartphone, networking, storage, and consumer markets. Key customers are Apple (10%+), Cisco, and Arris. The company manufactures its product in-house as well as at external foundries. Key competitors are Skyworks, Qorvo, Marvell, Cavium, and MicroSemi.

Broadcom (AVGO US) USD/(bn) unless otherwise noted

Income Statement	Oct FY	2024A	2025E	2026E	2027E	Q3/25A	Q4/25E	Balance Sheet	2024A	2025E	2026E	2027E
Revenue		51.6	63.5	88.2	113.9	16.0	17.6	Cash	9.3	19.2	59.8	114.0
EBITDA		40.7	50.2	65.2	81.6	12.7	13.6	Receivables	4.4	7.2	10.2	12.3
Depreciation		0.6	0.6	0.6	0.7	0.1	0.1	Inventories	1.8	2.4	3.1	3.5
Amortisation		9.4	8.1	6.7	5.4	2.1	1.9	Investments	0.0	0.0	0.0	0.0
EBIT		30.7	41.6	57.8	75.6	10.5	11.5	Fixed Assets	2.5	2.5	2.6	3.0
Net Interest Expense		3.4	2.7	2.2	2.2	0.7	0.5	Intangibles	138.5	130.2	123.5	118.2
Pre-Tax Profit		27.4	38.9	55.7	73.4	9.8	11.0	Other Assets	9.1	12.2	14.8	16.7
Tax Expense		3.6	5.4	8.4	11.0	1.4	1.5	Total Assets	165.6	173.6	214.1	267.7
Net Profit		23.7	33.5	47.3	62.4	8.4	9.5	Payables	1.7	1.6	2.1	2.3
Net Non Recurring & Assoc		0.0	0.0	0.0	0.0	0.0	0.0	Short Term Debt	1.3	1.4	1.4	1.4
Gross Cashflow		33.7	42.1	54.7	68.4	10.6	11.5	Long Term Debt	66.3	62.8	62.8	62.8
Diluted Shares Outstanding (m)		26.7	4.9	5.0	5.0	5.0	5.0	Provisions	2.0	1.7	1.7	1.7
EPS (adj) [\$]		2.9	6.8	9.5	12.6	1.7	1.9	Other Liabilities	26.8	26.2	30.2	32.5
CFPS (c)		4.0	8.5	11.0	13.8	2.1	2.3	Total Liabilities	98.0	93.7	98.2	100.7
Total DPS [\$]		1.1	2.3	2.3	2.3	0.6	0.6	Shareholders' Funds	67.5	79.7	115.7	166.7
								Minority Interests	0.0	0.0	0.0	0.0
								Other	0.2	0.2	0.2	0.2
								Total Shareholders' Equity	67.7	79.9	115.9	167.0
								Net Debt	58.2	45.0	4.4	(49.8)
								Net Debt / Equity (%)	86.0	56.4	3.8	(29.8)
								Interest Cover (x)	9.1	15.4	26.7	34.8
								ROE (%)	51.8	45.4	48.3	44.1
								ROA (%)	25.8	24.5	29.8	31.4
								ROIC (%)	54.4	28.4	39.4	53.4
Ratio		2024A	2025E	2026E	2027E			Cash Flow Analysis	2024A	2025E	2026E	2027E
Revenue Growth (%)		44.0	23.1	39.0	29.1	-	-	EBITDA	40.7	50.2	65.2	81.6
EBITDA Growth (%)		57.0	23.3	29.8	25.2	-	-	Chgs in Working Cap	(5.0)	4.0	1.9	1.8
EBIT Growth (%)		38.9	35.4	39.0	30.7	-	-	Other	(15.8)	(22.9)	(14.4)	(16.8)
EPS Growth (%)		582.7	134.7	40.6	31.9	-	-	Operating Cashflow	20.0	31.3	52.8	66.6
Total DPS growth (%)		0.5	0.1	0.0	0.0	-	-	Acquisitions	0.0	0.0	0.0	0.0
CFPS Growth (%)		0.7	0.1	0.0	0.0	-	-	Capex	(0.5)	(0.6)	(0.8)	(1.1)
EBITDA/Sales (%)		0.1	0.1	0.1	0.1	-	-	Other	(22.5)	0.2	0.0	0.0
EBIT/Sales (%)		0.1	0.1	0.1	0.1	-	-	Investing Cashflow	(23.1)	(0.4)	(0.8)	(1.1)
Payout Ratio (%)		38.3	33.4	24.0	18.2	-	-	Dividends Paid	(9.8)	(11.2)	(11.3)	(11.3)
EV/EBITDA (x)		44.6	36.7	28.3	22.6	-	-	Other	8.1	(9.9)	0.0	0.0
EV/EBIT (x)		59.2	44.3	31.9	24.4	-	-	Financing Cashflow	(1.7)	(21.1)	(11.3)	(11.3)
PGCFPS (x)		89.3	42.3	32.7	26.1	-	-	Net Chg in Cash/Debt	(4.8)	9.8	40.6	54.2
P/BV (x)		26.1	22.4	15.4	10.7	-	-	Free Cashflow	19.4	30.8	51.9	65.5
PE (adj) (x)		124.8	53.2	37.8	28.7	-	-					
Total Div Yield (%)		0.3	0.6	0.6	0.6	-	-					

Source: Company data, Macquarie Research Sep 2025

Executive summary

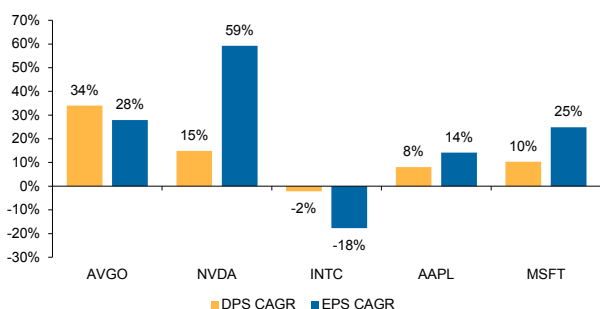
Key investment thesis

AI ASIC is growing faster than GPU, as detailed in our latest Ecosystem *AI Accelerator race* report. Broadcom is well positioned as the premier growth story in ASIC and infrastructure software, underpinned by management execution led by CEO Hock Tan, industry-leading SerDes IP, disciplined M&A, high-margin software expansion, and superior shareholder returns, including higher dividend growth and yield than peers. We initiate at Outperform, with an implied 17% total shareholder return to our US\$420 price target.

World-class management and operating strategy

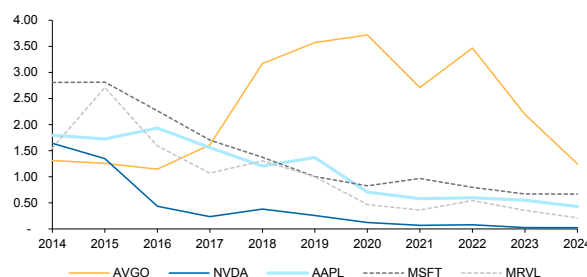
- **CEO leadership:** Broadcom's CEO Hock Tan's front-loaded, performance-based incentive plan aligns management with long-term shareholder value, ensuring executive continuity and strategic focus across Broadcom's multi-year M&A and growth roadmap.
- **Proven track record:** Under Tan's leadership, Broadcom has consistently delivered on synergy targets, margin expansion, and successful integration of major acquisitions (LSI, CA Technologies, Symantec, VMware).
- **Higher dividend growth and yield vs peers:** Broadcom has delivered a ~34% dividend CAGR over the past decade, outpacing both peers and earnings growth. It has raised its payout ratio from 21% in 2011 to about 64% in 2024, with management committing to returning ~50% of prior year free cash flow as dividends. Broadcom's dividend yield consistently ranks above industry averages.

Figure 1 - DPS and EPS CAGR in 2014-2024



Source: Factset, Macquarie Research, Sep 2025

Figure 2 - Dividend yield against peers

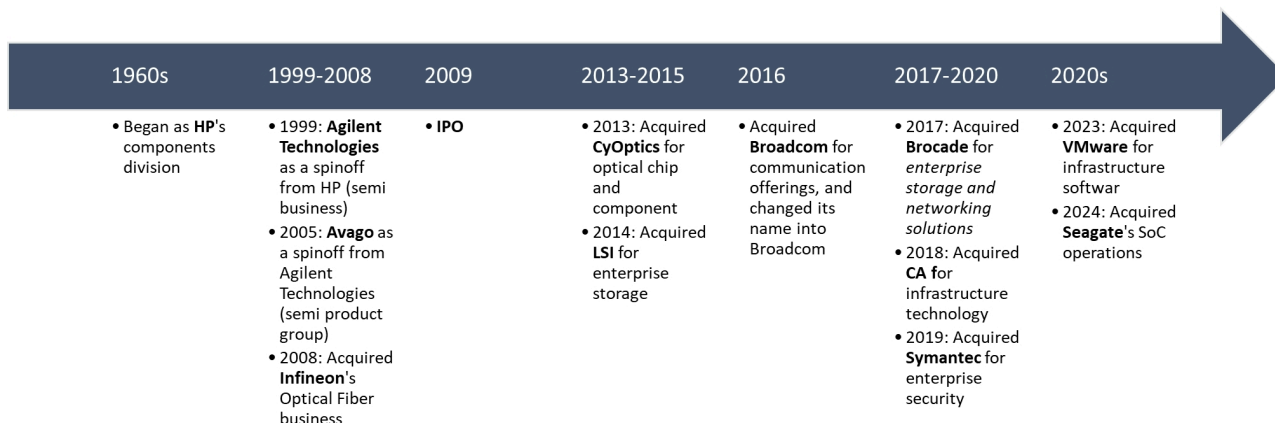


Source: Factset, Macquarie Research, Sep 2025

Disciplined, value-accretive M&A

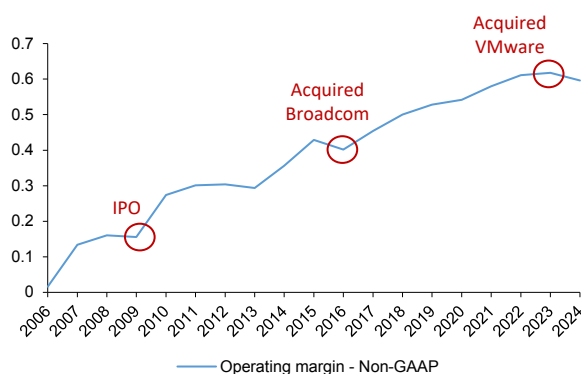
- **Proven integration track record:** Since 2015, Broadcom has executed over US\$100bn in acquisitions, consistently meeting or exceeding synergy and margin targets.
- **Strategic focus:** Management remains aligned, with a unique incentive plan designed to reinforce executive continuity and shareholder value creation.

Figure 3 - Key M&A Deals



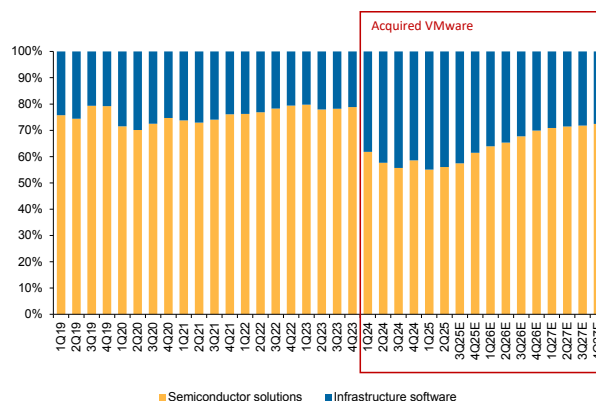
Source: Factset, Macquarie Research, Sep 2025

Figure 4 - Operating margin (Non-GAAP)



Source: Company data, Macquarie Research, Sep 2025

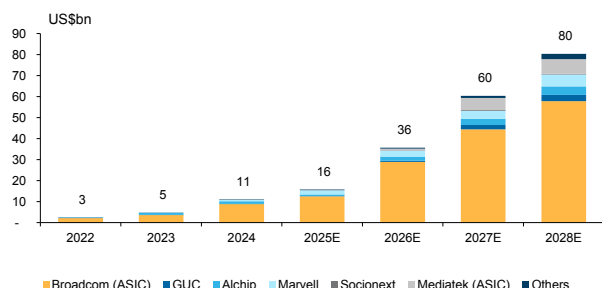
Figure 5 - Revenue Segment (USD'mn)



Source: Company data, Macquarie Research, Sep 2025

AI ASIC leadership

- **ASIC growth outpaces GPUs:** We project the global AI ASIC market is projected to deliver a 72% CAGR (2025–2028E), with Broadcom initially capturing over 70% market share. Its technology is expanding beyond hyperscalers into automotive, healthcare, and vertical AI markets.
- **Near-monopoly in AI networking:** Broadcom holds leading positions in Ethernet switching, routing, and SerDes IP, which are critical for scaling AI clusters. Increasing model complexity drives higher silicon content per rack, reinforcing Broadcom's dominance.

Figure 6 - ASIC market by key players (excl networking)

Source: Factset, Macquarie Research, Sep 2025

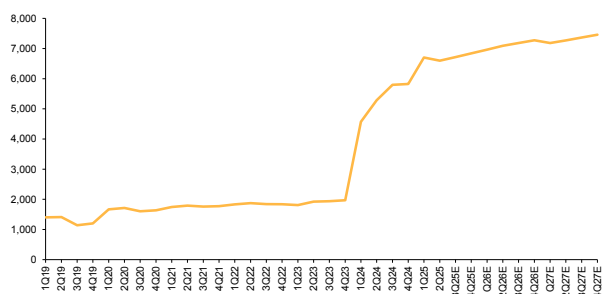
Figure 7 - Broadcom AI revenue (incl networking) sensitivity in FY27E

		FY27 SAM in US\$bn				
		50	60	75	90	100
AVGO	75%	38	45	56	68	75
	80%	40	48	60	72	80
Market	85%	42	51	63	76	85
	90%	45	54	68	81	90
Share	95%	48	57	71	86	95

Source: Company, Macquarie Research, Sep 2025

High-margin software expansion

- **VMware acquisition:** The transformative acquisition of VMware in 2023 has propelled software to over 40% of total revenue. Broadcom swiftly transitioned VMware's business model from perpetual licences to subscription, driving double-digit ARR growth and locking in high renewal rates.
- **Margin resilience:** Infrastructure software now contributes roughly 50% of EPS, with segment gross margins reaching 93%. The recurring nature of software revenue enhances long-term stability and free cash flow.

Figure 8 - Infrastructure software revenue estimates

Source: Company data, Macquarie Research, Jul 2025

Figure 9 - Broadcom software revenue sensitivity in FY27E

		FY27 TAM in US\$bn				
		1,300	1,500	1,600	1,700	1,900
AVGO	1.0%	13	15	16	17	19
	1.5%	20	23	24	26	29
Market	2.0%	26	30	32	34	38
	2.5%	33	38	40	43	48
Share	3.0%	39	45	48	51	57

Source: Company, IDC, Macquarie Research, Sep 2025

Figure 10 - Weighted average P/E

2027E	OP mix	PE
Semiconductor solutions	65%	23.3
Infrastructure software	35%	52.5
Weighted avg		33.5

Source: Company data, Factset, Macquarie Research, Sep 2025

Valuation and growth outlook

- **Premium justified by quality growth:** Our US\$420 target price is based on a blended 33.5x 2027E EPS multiple, reflecting strong sector growth, superior margins, and strategic execution.
- **Earnings growth:** Forecasted EPS growth of 30%+ annually through 2027, with upside potential from further software integration and AI demand.

Company overview

Broadcom is a global technology leader in connectivity semiconductor and infrastructure software solutions, with a focus on enabling high-performance computing, networking, and enterprise IT. The company is also known for its consistent and disciplined execution of an M&A-driven growth strategy. Broadcom has transformed itself over the past decade from a communication chip supplier into a diversified infrastructure platform spanning cloud, telecom, enterprise and consumer end markets.

Semiconductor solutions (~60% of revenue): Broadcom supplies custom and merchant silicon across data centre networking, AI accelerators, broadband access, wireless RF, storage, and industrial markets. Notable positions include:

- **AI networking and custom ASIC:** Leading switch (Tomahawk) and router (Jericho) chips for AI clusters; custom AI silicon for hyperscalers like Google and Meta.
- **Wireless:** Key suppliers of RF and connectivity modules to Apple, Samsung, and other smartphone brands, supported by long-term supply agreements.
- **Broadband & Storage:** Strong positions in DOCSIS (Data Over Cable Service Interface Specification), PCIe switching, and SSD controllers.

Infrastructure software (~40% of revenue): Built through acquisitions, this segment mainly includes:

- **VMware:** Virtualisation and private cloud software with high enterprise stickiness and cross-sell potential.
- **Symantec Enterprise:** Network and endpoint security for large enterprises.
- **CA Technologies:** Mainframe and DevOps software with long-term contracts and stable maintenance revenue.

Figure 11 - Revenue segment

Segment		Rev Contribution in FY24	Key Applications	Products	End Markets / Use
Networking		30%	Enables fast data transfer between computers, servers, and switches in large-scale networks	Ethernet switch/router chips (e.g. Tomahawk, Jericho) Custom ASIC Physical layer chips (PHYs) Fiber optic transceivers	Cloud companies (e.g. Google, Microsoft) Telecom operators Enterprises
Wireless		14%	Enables wireless communication in smartphone and tablets	WiFi/Bluetooth/GPS combo chips RF filters Power amplifiers Custom touch/charging controllers	Smartphone makers (e.g. Apple, Samsung) Wireless device manufacturers
Storage		7%	Helps servers and data centers store, manage, and access data	Storage controllers PCIe switches Fibre Channel adapters Ethernet server cards	Data centers Enterprise IT departments CSPs (e.g. Amazon, Meta)
Broadband		5%	Provide internet access at home and offices	Chips inside home routers, fiber-optic modems, set-top boxes (e.g. WiFi gateway SoCs, DSL/PON chips, DOCSIS modems)	Internet service providers (e.g. Comcast, China Mobile) Small businesses
Industrial		1%	Powers automation, sensing and communication in factories, vehicles, and medical equipment	Industrial sensors Optocouplers Motion control encoders Automotive Ethernet chips	Automotive Factory automation Green energy Healthcare equipment manufacturers
Infrastructure software (incl. VMware)		42%	Provides software for managing private clouds, enterprise security, and large IT systems	VMware Cloud platform Tanzu app management, SD-WAN Mainframe software Cybersecurity (Symantec, Arcot) Fibre Channel software	Large companies managing their own cloud or IT infrastructure (e.g. banks, hospitals, governments)

Source: Company data, Macquarie Research, Jun 2025

Networking

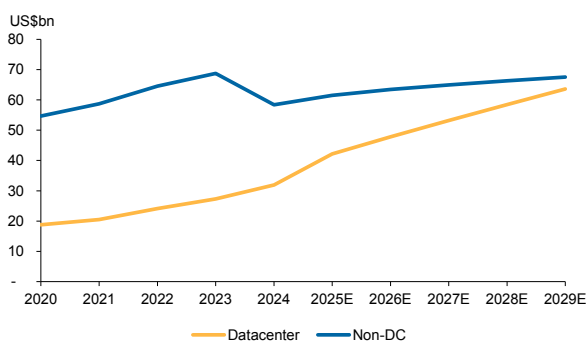
The AI era demands lightning-fast connectivity, and data centres are leading the charge in networking growth.

Broadcom positioned at the core of AI infrastructure growth

The global networking market is broadly divided into data centre (DC) and non-data centre (non-DC) segments. In the data centre context, networking infrastructure typically includes Ethernet switches, routing hardware, Fibre Channel (FC) switches, Infiniband switches, Application Delivery Controllers (ADC), ADC-as-a-Service (ADCaaS), and SDN controllers or network virtualisation overlays.

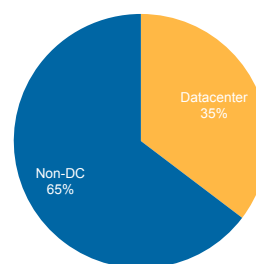
In 2024, the total networking market stands at US\$90bn according to IDC, of which data centre networking accounts for 35% (US\$32bn), while non-data centre environments (such as enterprise, campus, and edge networks) account for the remaining 65% (US\$58bn). However, the growth trajectory between these two segments diverges significantly. IDC projects a data-centre networking market CAGR of 15% over the next five years, to reach a scale of US\$64bn by 2029E. By contrast, it forecasts a non-data centre networking segment CAGR of just 3% over the same period, reaching US\$68bn by 2029E. This implies that nearly all net new growth in the networking market through 2029 will come from data centre deployments.

Figure 12 - Network by place



Source: IDC, Macquarie Research, Sep 2025

Figure 13 - Network by place (2024)



Source: IDC, Macquarie Research, Sep 2025

Ethernet switch is the largest and fastest-growing subsegment

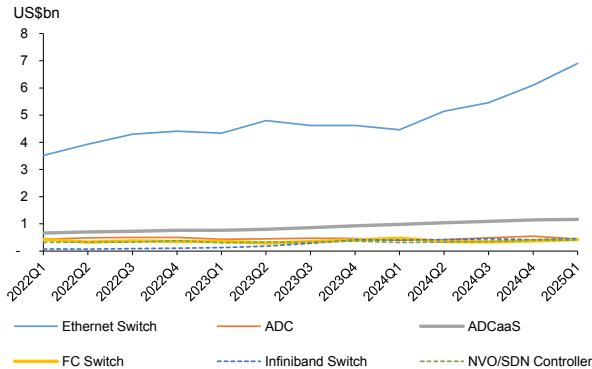
Within the data centre networking market, Ethernet switches represent the largest subsegment, accounting for ~66% of total data centre networking market in 2024. This category is also growing the fastest, fuelled by the intense east-west bandwidth demands of AI training, inference, and distributed storage architectures. The global data centre Ethernet switch market is valued at US\$21bn in 2024 as per IDC, growing at an 18% CAGR to reach US\$48bn by 2029E.

Traditional 100G links are being rapidly replaced by 200G and 400G, with 800G to scale meaningfully from 2025 onwards. By port speed, around 50% of the market today consists of 200G/400G deployments, while 100G still accounts for roughly 29%. The first wave of 800G-capable systems launched commercially in 1Q25, initially capturing around 5% of the market. We expect the adoption of 800G to accelerate as hyperscalers and AI infrastructure providers upgrade their networks to accommodate larger training clusters and higher interconnect densities.

The US represents the largest and most dynamic market for data centre networking, driven by concentrated AI capex from cloud providers such as Meta, Microsoft, Amazon and Google. US hyperscalers are typically first to adopt new port speeds and networking standards, making them the leading consumers of 800G and early adopters of 1.6T infrastructure. China also plays a significant role in networking demand, supported by domestic cloud service providers and national AI infrastructure build-outs, though its growth is partially constrained by US export restrictions and vendor substitution dynamics. In Europe, ongoing investments in sovereign cloud infrastructure, enterprise digitisation, and

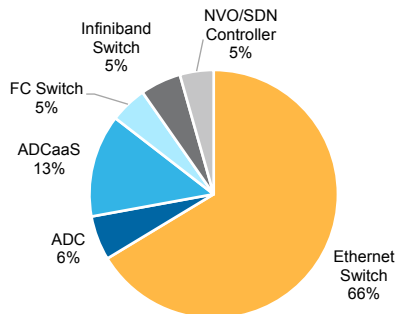
telco-to-cloud transformation will support steady demand, even though at a slower pace than in the US.

Figure 14 - Data centre networking market by technology



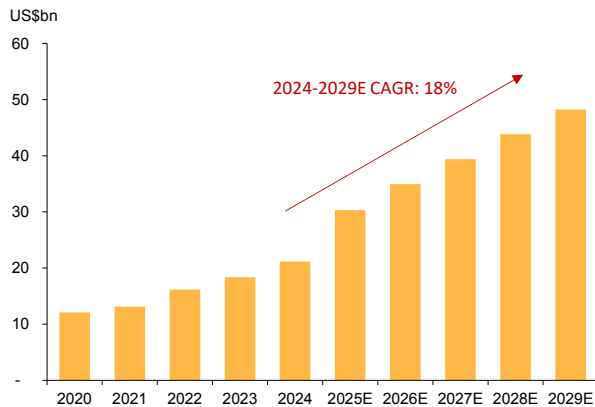
Source: IDC, Macquarie Research, Sep 2025

Figure 15 - Data centre networking market by technology (2024)



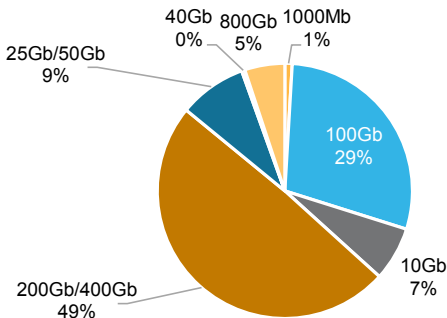
Source: IDC, Macquarie Research, Sep 2025

Figure 16 - Data Centre Ethernet Switch Market



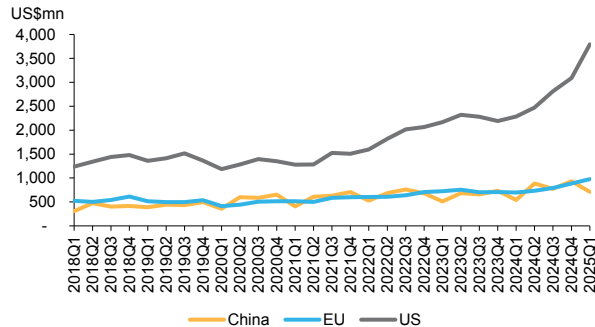
Source: IDC, Macquarie Research, Sep 2025

Figure 17 - Data Centre Ethernet Switch Market by speed (1Q25)



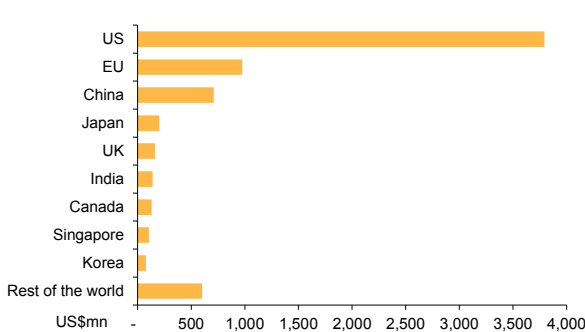
Source: IDC, Macquarie Research, Sep 2025

Figure 18 - Data Centre Ethernet Switch Market by geography



Source: IDC, Macquarie Research, Sep 2025

Figure 19 - Data Centre Ethernet Switch Market by geography (1Q25)



Source: IDC, Macquarie Research, Sep 2025

Broadcom's strategic role in the networking stack

Broadcom is a foundational supplier of the core silicon and connectivity components that power modern data centre networking infrastructure. Its product portfolio includes merchant Ethernet switch and router silicon, custom ASICs for hyperscalers, optical and copper physical layer (PHY) devices, and fibre-optic transceivers used in pluggable optical modules. These products are critical to enabling the scale-out, high-bandwidth networks required to support generative AI training clusters and cloud-scale computing.

Currently, about 70% of company's networking revenue is derived from AI data centre applications. The remaining 30% is linked to traditional enterprise and telco networking. Within its data-centre business, Broadcom supports both merchant silicon deployments through OEM partners and custom silicon engagements with hyperscalers, offering tailored architectures optimized for proprietary AI workloads.

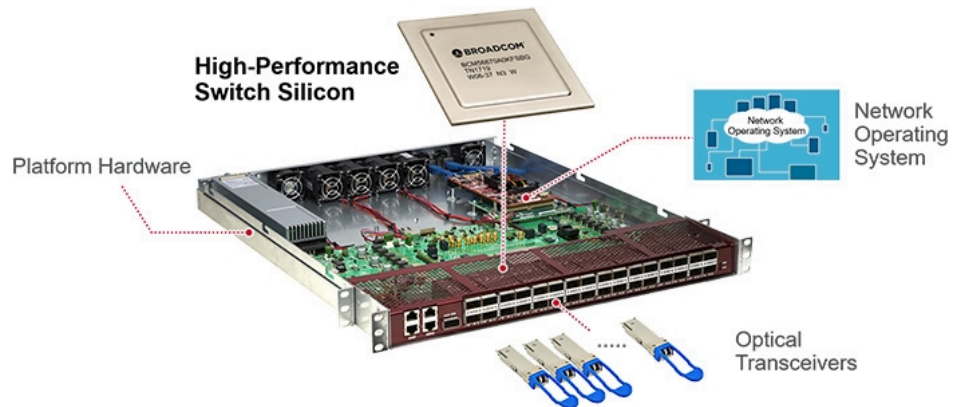
Networking (Tomahawk, Jericho, Ethernet/AI switches) is the largest and fastest-growing line of the company. Broadcom commands ~80% share of the global merchant Ethernet switch ASIC market, with Tomahawk and Jericho families setting the industry standard for AI data centre connectivity. Management reiterated its expectation that the serviceable addressable market (SAM) for XPU and AI networking could reach US\$60-90bn by 2027. AI networking now represents 40% of AI segment revenue, but management expects it to fall below 30% as custom ASIC ramps up.

- **AI infrastructure build-outs are the primary growth driver.** Networking chips for AI accounted for 70% of Broadcom's US\$2.5 billion 2Q25 networking sales, with AI connectivity revenue up 2x YoY.
- **Hyperscaler capex is at record highs.** 1Q25 spend reached US\$92bn, with 2025 AI/networking capex forecast to grow 44% YoY to US\$414bn. Broadcom's content share is expanding as Ethernet gains ground over InfiniBand in AI clusters. Product roadmap is tightly linked to hyperscaler AI cluster bandwidth needs. Tomahawk 7/8 and co-packaged optics are in development, with a path to 3.2T switching.
- **Tomahawk and Jericho** are the backbone of hyperscaler AI clusters, enabling 400G/800G/1.6T fabrics and supporting clusters of 100,000+ accelerators in two-tier topologies. Tomahawk 6 (102.4T) launches in 2H24, maintaining Broadcom's 1-1.5 year technology lead over merchant competitors. No merchant vendor approaches Broadcom's scale in high-end (400G/800G/1.6T) switch silicon.
- **Competitive edge** is reinforced by hyperscaler design wins, deep software/hardware co-design, and a multi-year lead in high-radix, low-latency switching.

Broadcom's deep engagement in Ethernet switching and custom ASIC places it at the centre of the ongoing shift toward data centre-centric networking. As AI training clusters scale in size and complexity, the demand for high-throughput, low-latency, and power-efficient interconnects is increasing rapidly. This trend aligns directly with Broadcom's core competencies in switch silicon, PHY integration, and pluggable optics.

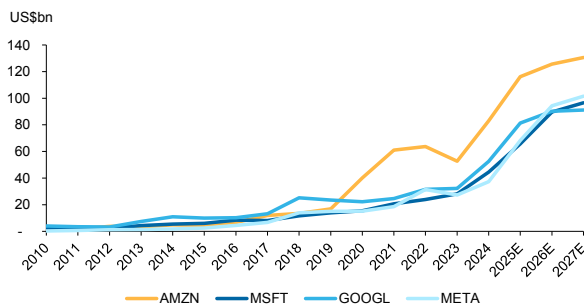
The company's ability to serve both the merchant silicon market and the high-value custom ASIC programs with hyperscalers provides a dual revenue stream with high visibility and strong operating leverage. With Ethernet-based AI clusters gaining share over proprietary fabrics (especially outside Nvidia-dominated environments), Broadcom is structurally positioned to benefit from both the expansion and architectural evolution of AI data centres.

Figure 20 - Server switch board



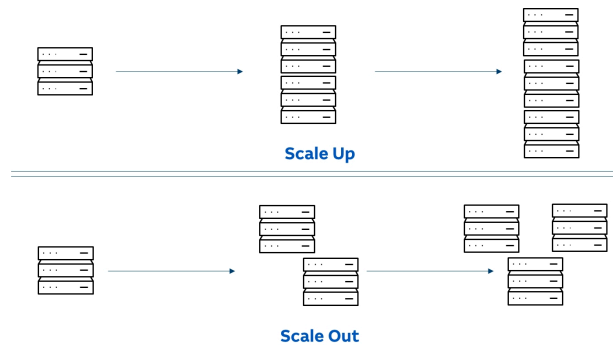
Source: Broadcom, September 2025

Figure 21 - Capex for major CSPs



Source: Factset, Macquarie Research, Sep 2025

Figure 22 - Scale up vs scale out



Source: Macquarie Research, Sep 2025

SerDes is core to company's moat

SerDes (Serialiser/Deserialiser) is one of the most technically demanding elements in semiconductor design, requiring mastery over high-speed signal integrity challenges, such as crosstalk, attenuation, and noise, while balancing analogue design with powerful digital signal processing (DSP).

Why SerDes is hard?

- **Extreme signal integrity and physical limitation.** SerDes must transmit extremely high-speed serial data (e.g., 112 Gb/s or 224 Gb/s per lane) over imperfect mediums like copper traces, PCB layers, or backplanes. As speeds escalate, maintaining a reliable signal over distance becomes exponentially more complex. Variability in materials and manufacturing processes further complicates design margins and consistency.
- **Complex analogue and digital co-design.** SerDes circuits require a sophisticated blend of analogue front-end design and advanced DSP (equalisation, clock-data recovery, re-timing). Crafting stable, low-power, high-bandwidth analogue cores paired with powerful DSP logic is highly specialised, demanding tight integration, low latency, and energy efficiency. Achieving this scale requires deep analogue-DSP expertise built up over years of design and manufacturing experience.
- **Lengthy development cycles and custom validation.** Building high-performance SerDes IP often takes multiple years, with each design validated across numerous configurations under real-world conditions - a resource-intensive process rarely feasible for smaller or less specialised firms.

- **Integrating a SerDes block into an ASIC chip presents another significant challenge.**

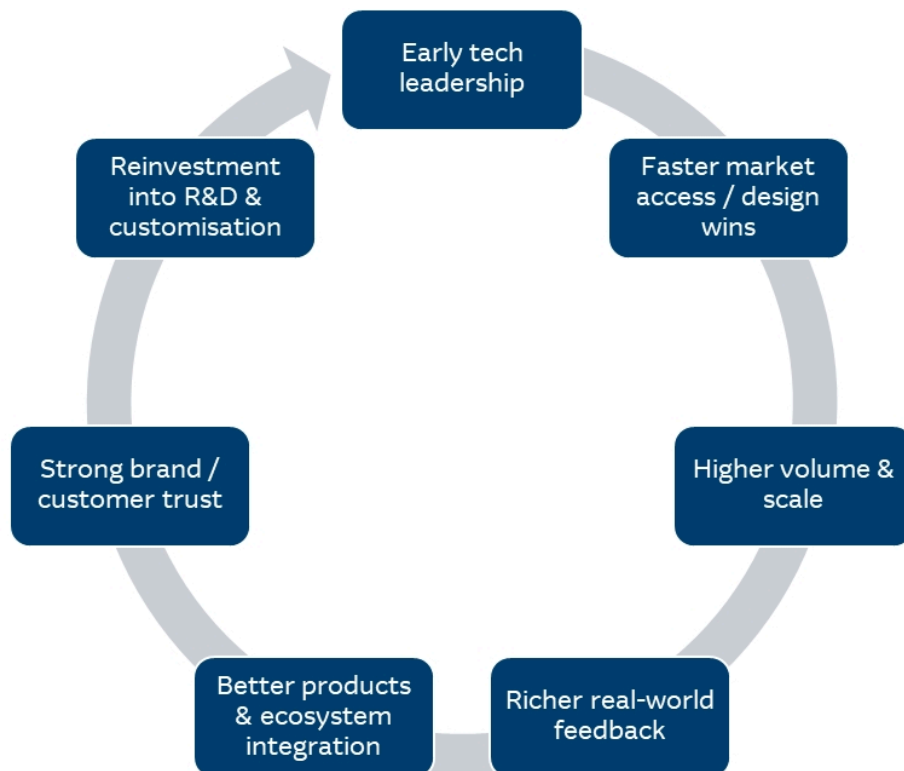
For chip-to-chip interconnects, both the switch and the ASIC require SerDes blocks. However, the ASIC's die area is already heavily populated with computing components, making it difficult to allocate space for additional SerDes circuitry.

The challenge is so acute that in 2025 Arm explored acquiring Alphawave to secure high-speed SerDes IP, while Qualcomm later announced a US\$2.4 billion deal to acquire Alphawave to strengthen its data-centre portfolio. Broadcom's position reflects years of IP investment, proven performance, and trusted integration, creating a durable moat that few competitors can penetrate.

First-mover track record. Broadcom has consistently launched next-generation SerDes and ASIC solutions ahead of competitors, giving it a "first-mover" edge. Early market entries build confidence amongst hyperscalers and large OEMs, enabling Broadcom to refine designs through real-world feedback and accelerate innovation.

- In June 2025, Broadcom announced the Tomahawk 6, the world's first 102.4Tb/s switch, adding 200G-per-lane SerDes options with longest-reach passive copper support, evidence of continued first-to-market execution at the next speed tier.
- The Tomahawk 5 switch chip incorporates 512×100G PAM4 SerDes, one of the industry's most flexible, long-reach 100G implementations, offering unmatched performance and integration.
- Complementing the fixed-switch cadence, Jericho 4 (Aug 2025) delivers 200G PAM4 SerDes on a 3 nm node with reach sufficient to eliminate retimers and streamlining deployment for hyperscalers.

Deep collaborative ecosystem. Broadcom works closely with hyperscalers in multi-year co-engineering partnerships, providing design, manufacturing, and integration of key IP such as SerDes and memory subsystems (e.g., HBM integration via its 3.5D XDSiP platform). This enables customers to scale custom chips without maintaining full in-house semiconductor teams. Broadcom's custom-ASIC partnerships are widely reported to include Google and Meta; Reuters has also reported collaborations with ByteDance (see news [here](#)) and OpenAI (see news [here](#)). We believe that this roster is will expand as demand for advanced SerDes IP rises, giving Broadcom a unique selling point in ASIC programs. The vertically integrated model strengthens both product appeal and customer retention, long validation cycles and high integration complexity create substantial switching costs. Once embedded in infrastructure, Broadcom's IP is difficult to replace, reinforcing a virtuous cycle of innovation and market share gains.

Figure 23 - A virtuous cycle for first-mover advantage

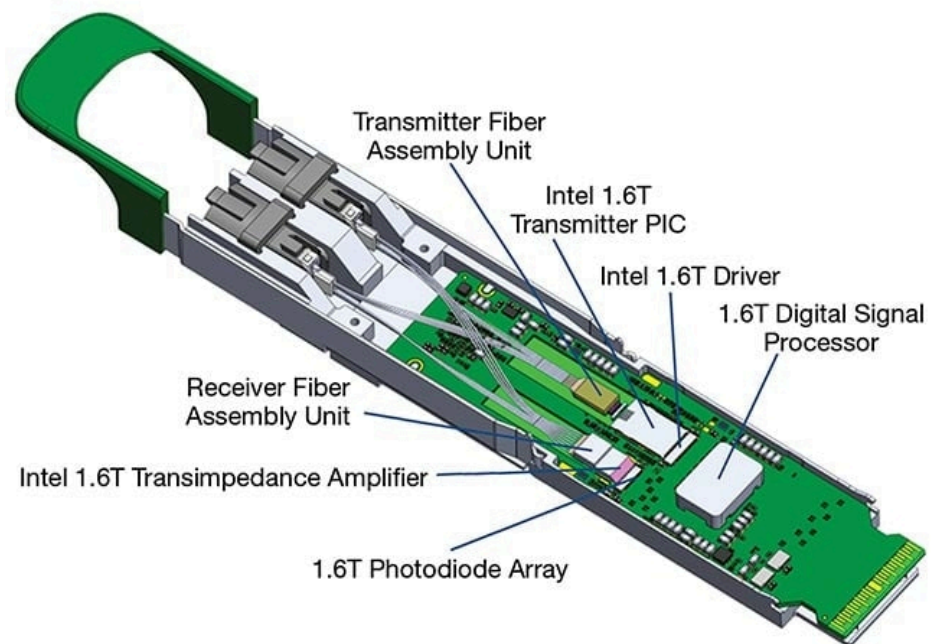
Source: Macquarie Research, Sep 2025

Co-Packaged Optics (CPO) is revolutionising networking by seamlessly integrating optics with computing dies, unlocking unprecedented speed and efficiency.

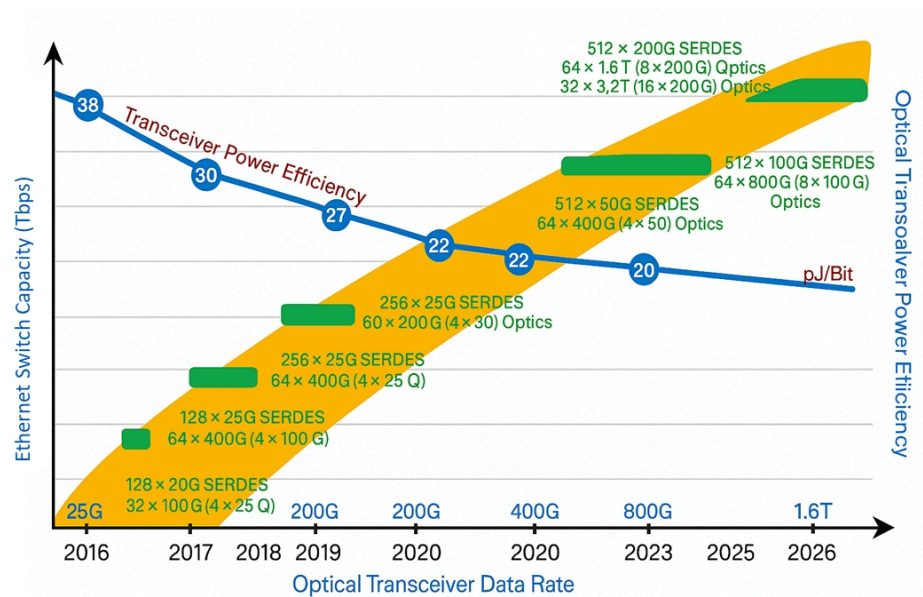
Smaller, faster, cooler: CPO reshapes networking from plug-in to built-in

Co-packaged optics (CPO) tightly integrates optical transceivers (silicon photonics engines) with switch ICs on the same package or board. Instead of pluggable modules on the faceplate, CPO bundles optical transceivers closer to the chips, significantly shortening the electrical path. This architecture enables much higher aggregate bandwidth, lower latency, and far better energy efficiency compared to traditional optics. CPO can reduce power per bit by ~70% compared to pluggable solutions (see news [here](#)), and achieve 10x better signal integrity by shortening data travel length. These gains are critical in AI/HPC data centres where network bandwidth doubles every few years. CPO is the next wave of data-centre switching, giving hyperscalers the power and density needed for large AI clusters.

Broadcom is a clear leader in CPO, with dedicated switch and photonic solutions and hyperscaler partnerships. Its flagship products are the StrataXGS Tomahawk series with co-packaged optical engines. In 2022, Broadcom and Tencent announced a 25.6Tb/s CPO switch ("Humboldt") based on Tomahawk 4 with four 3.2Tb/s photonic chiplets. In 2024, Broadcom delivered "Bailly", the industry's first 51.2Tb/s CPO switch, which combines a Tomahawk 5 with eight 6.4Tb/s silicon-photonics engines. This integration yields an optical interconnect consuming 70% less power and 8x higher silicon efficiency than a pluggable-optics design.

Figure 24 - An architecture of a 1.6T module

Source: Photonics, Intel

Figure 25 - Ethernet switch and optical transceiver trend

Source: Photonics, Intel

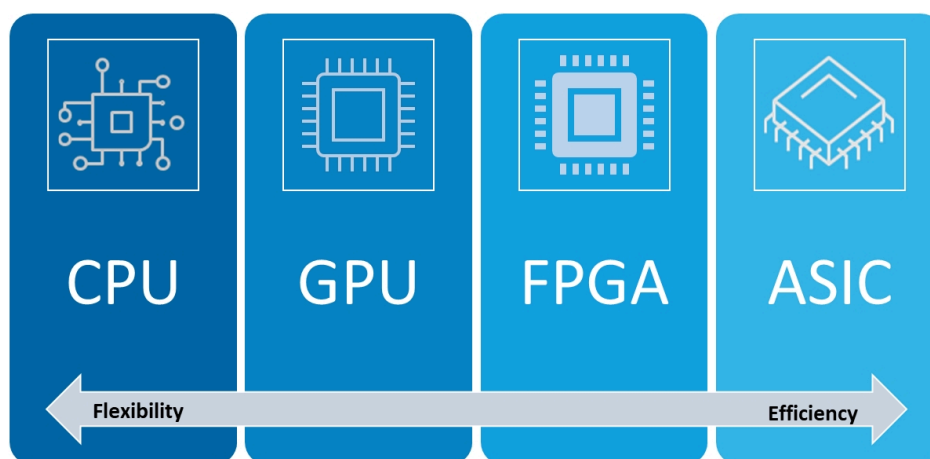
Custom ASICs are set to outpace GPUs, driven by the explosive growth of inferencing workloads that demand tailored, high-efficiency solutions

Custom ASICs to outgrow GPUs

Structural shift favours the ASIC supply chain

GPUs, such as Nvidia's B200, are designed for broad-purpose parallel compute and are widely used for AI training. They are flexible and programmable, which makes them ideal for model experimentation and for customers who need off-the-shelf acceleration without custom hardware development. However, this flexibility comes with trade-offs, particularly in terms of power efficiency, unit cost, and I/O bottlenecks in large-scale deployments. In contrast, ASIC are custom chips designed for a specific workload. In AI infrastructure, these include inference accelerators (e.g. Google TPU), training chips, or networking ASIC tailored to a hyperscaler's unique model architecture or system design. While ASIC requires longer design cycles and significant up-front investment, they offer unmatched performance per watt, system-level efficiency, and tighter integration across the compute rack.

Figure 26 - Chip types



Source: Maquarie Research, Sep 2025

ASIC will grow faster than GPU in inference and vertical AI deployments

Inference volume favours ASIC. As AI models move into production at scale, inference workloads vastly outnumber training workloads. Unlike training, which is often centralised and occasional, inference is deployed across global user-facing systems (e.g. search, recommendations, video encoding, ad targeting). ASICs are better suited for inference because of:

- **High energy efficiency**, which leads to lower TCO per inference.
- **Lower latency**, which is important for real-time applications.
- **Customisable** for quantised models and optimised dataflows.

Google's TPU and Amazon's Inferentia are already replacing GPUs for internal inference workloads at scale. As more hyperscalers roll out proprietary models, ASIC will become the default choice for inference deployment.

Vertical integration and cost optimisation. Hyperscalers increasingly seek vertical control over AI infrastructure to reduce dependence on Nvidia, mitigate supply constraints, and improve cost-efficiency. Custom ASICs can:

- Mitigate Nvidia's high markup and bundled software lock-in
- Allow system-level co-design with networking, memory, and cooling
- Are designed to exactly match model architecture, reducing over-provisioning

As large-scale LLM and recommender systems mature, custom ASIC deployments will grow faster than GPU upgrades, especially for stable repeatable production workloads.

Figure 27 - AI accelerator market share by chip type

	2023	2025E	2027E
GPU	72%	65%	62%
ASIC	22%	30%	31%
CPU	4%	3%	4%
FPGA	2%	2%	3%

Source: The Information Network, Macquarie Research, Sep 2025

Figure 28 - Market share of GPU vs ASIC chips by training and inference

	2023	2025E	2027E
Training			
GPU	90%	87%	83%
ASIC	10%	13%	17%
Inference			
GPU	45%	40%	38%
ASIC	55%	60%	62%

Source: The Information Network, Macquarie Research, Sep 2025

ASIC adoption is broadening beyond hyperscalers

While CSPs were early ASIC adopters, the trend is expanding to consumer tech, automotive, and vertical AI players, as companies seek control over performance, cost, and system architecture. Below are key examples.

- **Google's** TPU series is the most mature custom AI silicon program in the industry, which is optimised for inference and scaled-out AI workloads at high efficiency. Google has also developed its own interconnects but still relies heavily on Broadcom for cloud infrastructure. Broadcom has long been a custom silicon partner to Google for both networking and ASIC components.
- **Apple** is reportedly (news [here](#)) working with Broadcom to develop a custom AI server chip to power its internal gen-AI infrastructure. The chip is designed to support AI inference workloads behind Siri, iOS features, and iCloud AI processing, complementing Apple's on-device Neural Engine.
- **Meta's** MTIA chip is built to optimise AI inference for ranking, recommendation, and feed personalisation. Meta expects to develop MTIA deployments and reduce dependence on Nvidia, especially for lower-precision inference use cases. Broadcom is a key supplier of custom Ethernet switch and interconnect silicon in MTIA-powered AI clusters.
- **Tesla** is building its own AI training supercomputer (Dojo), based on a fully custom ASIC designed in-house. The chip is optimised for video-based AI workloads used in autonomous driving and FSD training. While Tesla uses proprietary logic, Broadcom remains a key vendor in the networking and high-speed interconnect layers.
- **Amazon's** AWS Inferentia and Trainium chips are purpose-built to reduce reliance on Nvidia for both inference and training. AWS is vertically integrating silicon into its AI service stack.
- **Chinese CSPs (eg, Baidu Kunlun, Alibaba Hanguang, Tencent Zixiao).** Facing US export controls, Chinese hyperscalers are accelerating internal ASIC development. Baidu's Kunlun and Alibaba's Hanguang are deployed in domestic cloud AI services and public LLM offerings. These chips prioritise inference at scale and integrate with home-grown frameworks.

Figure 29 - Key ASICs in the market

Company	Chip	Process	Design service provider	Mass production
AWS				
	Inferentia 2	7nm TSMC	Alchip (back-end)	2H22
	Inferentia 2.5 / Trainium 2	5nm TSMC	Marvell (back-end)	2H24
	Inferentia 3 / Trainium 3	3nm TSMC	Alchip (back-end)	2H25/1H26
	Graviton v3 (CPU)	5nm TSMC	AWS in-house	2022
	Graviton v4 (CPU)	4nm TSMC	AWS in-house	2024
Google				
	TPUv5p	7nm TSMC	Broadcom	2023
	TPUv6e (Trillium)	5nm TSMC	Broadcom	2024
	TPUv6p	3nm TSMC	Broadcom	2025
	TPUv7	3nm TSMC	Mediatek (back-end)	2H26
	Axion (CPU)	5nm TSMC	Marvell (back-end)	1H24
	Axion 2 (CPU)	3nm TSMC	GUC (wafer logistics mgnt)	2026
MSFT				
	Maia 100	5nm TSMC	GUC (wafer logistics mgnt)	1H24
	Maia 200	3nm TSMC	Marvell (back-end)	2026
	Cobalt 100 (CPU)	5nm TSMC	GUC (wafer logistics mgnt)	2H24
	Cobalt 200 (CPU)	3nm TSMC	GUC (wafer logistics mgnt)	2026
Meta				
	MTIA	7nm TSMC	Broadcom	2H23-1H24
	MTIA v2	5nm TSMC	Broadcom	2H25
	MTIA v3/MTIA T-V1	3nm TSMC	Broadcom	2H26
	(CPU gen 1)	3nm TSMC	SocioNext	2026
Intel (Habana)				
	Gaudi 2	7nm TSMC	Alchip (back-end)	2H23
	Gaudi 3	5nm TSMC	Alchip (back-end)	2H24
	Jaguar Shores			
Tesla				
	FSD2	5nm Samsung	Broadcom	2H23
	Dojo D2	5nm TSMC	Broadcom	Late 2025
	Dojo D3			Late 2026
Apple				
	Baltra	3nm TSMC	Broadcom	2026
Open AI				
	Self-developed AI chip			1Q25

Source: Macquarie Research, Sep 2025

Nvidia faces limitation at scale, and Broadcom is a prime beneficiary

Nvidia's leadership in AI compute is clear, but its model is increasingly constrained at the infrastructure level. The following factors open the door for custom ASIC players and make ASIC-based architectures more attractive for cost and scalability.

- **High TCO (total cost of ownership):** Nvidia solutions are expensive due to bundled software, high ASPs, and integrated networking (e.g. NVLink, NVSwitch, Infiniband)
- **Interconnect lock-in:** Nvidia's proprietary stack limits flexibility and interoperability.
- **Supply chain bottleneck:** As demand for Blackwell chip surges, hyperscalers are incentivised to pursue alternative, internal paths.

Broadcom is uniquely positioned in the ASIC supply chain, offering both merchant and custom silicon solutions tailored to AI infrastructure.

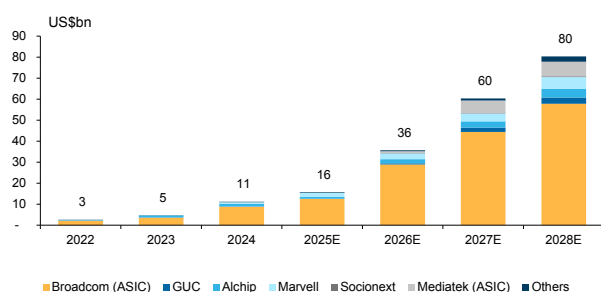
- **Custom AI ASIC:** Broadcom is a key supplier of AI accelerator ASICs for major hyperscalers, including Google and Meta. These chips are developed jointly with customers and tightly integrated into their software stacks.
- **Networking leadership:** Broadcom's Tomahawk, Jericho, and custom Ethernet switch chips are foundational to AI clusters that use Ethernet fabric. The growth is accelerating amongst non-Nvidia environments.
- **Vertical integration:** Broadcom also supplies SerDes, PHYs, and optical DSPs that tie together high-speed networking fabrics, enabling it to capture value across the ASIC-to-optics stack.

- **Scalable operation model:** With its fabless model and centralised engineering approach, Broadcom can support multiple custom silicon programs without bloating R&D. This enables higher operating leverage as the ASIC market scales.

ASIC market outlook

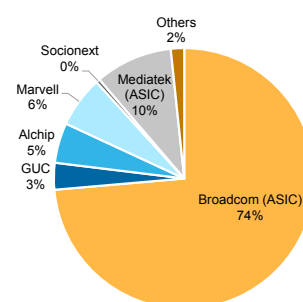
We project a 72% ASIC market CAGR over 2025-28E, growing from sub-US\$11bn in 2024 to over US\$80bn by 2028 (excluding networking). Broadcom itself guides for a US\$75bn SAM by 2027 (including networking). We estimate that networking accounts for ~20% of the total ASIC serviceable market. We believe Broadcom is well positioned to capitalise on this rapid growth, sustaining its market leadership with an estimated ~75% share of the ASIC market by 2027. While smaller entrants such as MediaTek are beginning to emerge, we see Broadcom's scale, track record, and customer relationships as key differentiators that should allow it to defend market leadership.

Figure 30 - ASIC market by key players (excl networking)



Source: Factset, Macquarie Research, Sep 2025

Figure 31 - 2027 ASIC market by key players



Source: Factset, Macquarie Research, Sep 2025

To illustrate the range of outcomes, we provide a sensitivity analysis of Broadcom's potential AI ASIC revenue in FY27E based on different assumptions for total SAM and Broadcom's market share. Under our base case (i.e. US\$75bn SAM, 85% share), AI revenue would reach ~US\$63bn. However, outcomes range from ~US\$38bn (i.e. US\$50bn SAM, 75% share) to as high as ~US\$95bn (US\$100bn SAM, 95% share). This highlights both the robustness of Broadcom's positioning and the significant upside potential if SAM expands faster and Broadcom consolidates share.

We extend the AI ASIC sensitivity to EPS. In our base case (US\$75bn SAM, 85% share), we forecast EPS of US\$12.6 in 2027. A downside case (US\$50bn SAM, 75% share) implies EPS of US\$9.9, while an upside case (US\$100bn SAM, 95% share) could lift EPS to US\$15.7. This analysis underscores both the downside protection from Broadcom's existing businesses and the meaningful EPS leverage to AI ASIC growth.

Figure 32 - Broadcom AI revenue (incl networking) sensitivity in FY27E

		FY27 SAM in US\$bn				
		50	60	75	90	100
AVGO	75%	38	45	56	68	75
	80%	40	48	60	72	80
	85%	42	51	63	76	85
	90%	45	54	68	81	90
	95%	48	57	71	86	95

Source: Company, Macquarie Research, Sep 2025

Figure 33 - Broadcom EPS sensitivity in FY27E

		FY27 SAM in US\$bn				
		50	60	75	90	100
AVGO	75%	9.9	10.7	11.8	13.0	13.7
	80%	10.2	11.0	12.2	13.4	14.2
	85%	10.4	11.3	12.6	13.8	14.7
	90%	10.7	11.6	13.0	14.3	15.2
	95%	10.9	11.9	13.3	14.8	15.7

Source: Company, Macquarie Research, Sep 2025

Other semiconductor products, long accustomed to cyclical volatility, are now stepping out of a downturn

Other semiconductor solutions

Wireless (Smartphone components)

Wireless products enable wireless communication in smartphone and tablets, include WiFi/Bluetooth/GPS combo chips, RF filters, power amplifiers and custom/touch/charging controllers. This segment is heavily tied to its key smartphone customers (e.g. Apple), which has been a consistent but more mature revenue generator. Performance has been closely linked to product cycles of major smartphone launches. In 4Q24, wireless jumped to US \$2.2bn (7% YoY and 30% QoQ) amid a strong iPhone launch. Management noted that Apple accounts for essentially most of its wireless sales. iPhone 5G and RF content upgrades drove wireless revenue spikes in 2020-2022, but lacked sustained elasticity. Wireless plateaued at US\$7.3bn in FY23, with no material new design wins or content step-ups. Tariff and market risks (e.g. US-China trade affecting Apple) have been noted and thus far Broadcom's wireless revenue is seeing modest to flat growth. Key competitors include Qualcomm, which retains a dominant share in Android RF and modems. ASP is stable to down, with incremental RF content gains offset by flat iPhone unit sales and limited new design wins. We estimate that annual price increases would fail to offset cost inflation, and gross margin will trend lower as mix shifts towards lower incremental content. We estimate that the segment market now lags AI/networking.

Broadband (Fixed infrastructures)

The broadband segment (e.g. cable modems, Passive Optical Network or PON, Digital Subscriber Line or DSL, and fixed-wireless gateways) has weakened since COVID. Broadband posted strong growth driven by the surge in work-from-home arrangements during the pandemic, as well as global fibre rollouts and WiFi 6/6E upgrades. DOCSIS 3.1/4.0 (i.e. data over cable service interface specification, which is a tech that delivers internet over cable TV lines) and 10G PON adoption accelerated in 2022. However, Broadband revenue peaked at US\$4.5bn in FY23, up 8% YoY, but Q4 revenue declined by 9% YoY as service provider inventory digestion began. The segment hit a cyclical bottom in 4Q24 at US\$465mn, down 51% YoY, as telco and cable customers paused orders to work through excess inventory. Broadcom management noted that a gradual recovery in FY25. 2Q25 broadband revenue showed sequential growth and it expects it to continue growing sequentially in 3Q25. As enterprise/telco semiconductor spend is cyclical, we expect broadband revenue to remain under pressure until carriers resume rollouts.

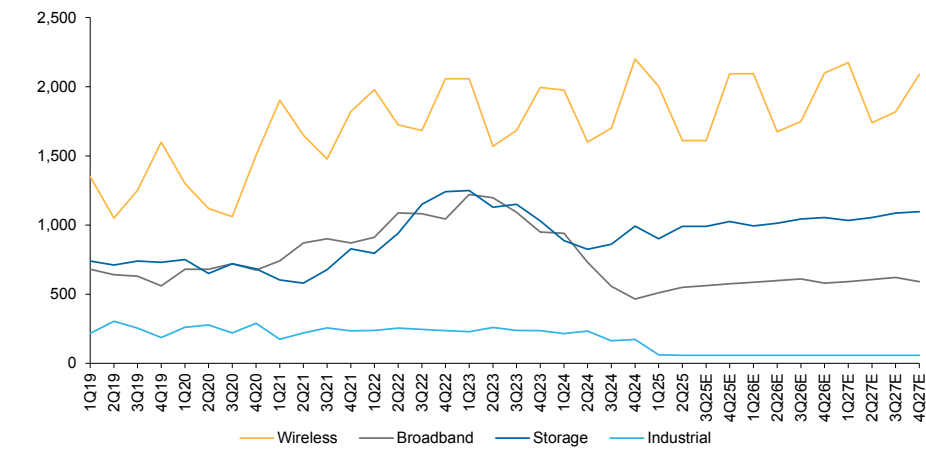
Storage (Server & enterprise storage)

Broadcom's storage connectivity (e.g. SATA controllers for consumer PCs/laptops, SAS controllers for servers and data centres, SSD controllers for NAND flash memory, etc.) saw a downturn in FY23-FY24, reflecting subdued data-centre buying outside hyperscalers. Hyperscalers' demand for NVMe/HPC drives and custom HBAs provided some offset, but overall enterprise spend was weak. GPM in storage are high (controller ASICs). Broadcom traditionally wins major OEM server deals, so, while its share is stable, total demand has fallen.

Industrial (Diversified semiconductors)

The firm's industrial segment mainly includes mixed-signal chips, factory automation, automotive, and IoT connectivity chips. 4Q24 saw a larger drop as broad technology spending softened. Broadcom serves industrial markets through diversified analogue/mixed signal offerings. Growth drivers will include 5G IoT modules and factory automation chips, but spending in these markets have only recovered slowly post-COVID. Overall, industrial remains a small (1-2% of revenue), albeit steady contributor with modest growth.

Figure 34 - Other revenue estimates



Source: Company data, Macquarie Research, Jul 2025

Infrastructure software

The acquisition of VMware has propelled software revenue to a commanding 40% of total revenue.

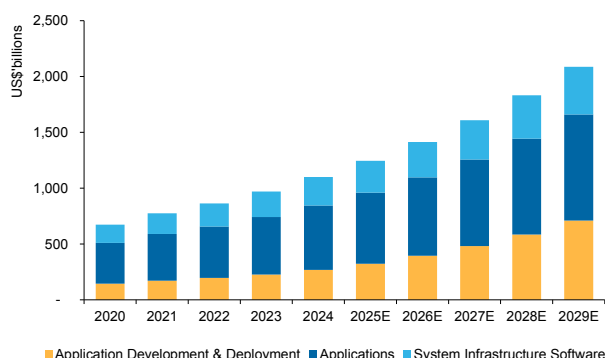
Infrastructure software refers to the foundational layer of enterprise IT systems that enables the functioning, management, and security of hardware, networks, and applications. It includes operating systems, virtualisation platforms, middleware, system management tools, and cybersecurity solutions. These components are essential for running mission-critical workloads in data centres, cloud environments, and increasingly, hybrid enterprise infrastructure.

According to IDC, the infrastructure software market will grow from US\$1.1tn in 2024 to US\$2.1tn in 2029, representing an 11% CAGR. We estimate that Broadcom's software revenue will increase from US\$21.5bn in 2024 to US\$32.2bn by 2027, a 15% CAGR. The company's software exposure is highly skewed toward the infrastructure software segment, which includes mission-critical domains such as virtualisation (eg, VMware), security (eg, Symantec), and operations management (eg, CA Technologies). The acquisition of VMware, which closed in late 2023, has fundamentally transformed Broadcom's infrastructure software segment, strengthening its revenue contribution and strategic importance.

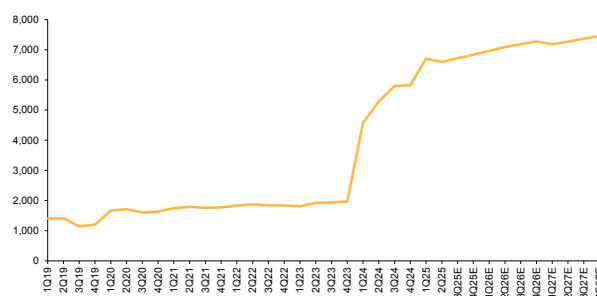
- **Prior to VMware acquisition,** the infrastructure software segment, built on the acquisitions of CA Technologies and Symantec's enterprise security business, delivered stable, recurring revenue. The focus was on a portfolio of mainframe and cybersecurity software. Subscription/maintenance revenue dominated, with renewal rates averaged 120% over expiring contracts.
- **Upon the integration of VMware,** Broadcom moved swiftly to restructure VMware's business model, shifting from perpetual licences to a subscription-based model. This led to a significant and immediate uplift in the segment's reported revenue in FY24 into FY25.
 - ⇒ **Subscription model transition:** The move to a subscription model for VMware's portfolio, particularly VMware Cloud Foundation (VCF), has been a central pillar of Broadcom's strategy. While initially met with some customer concern, management has reported strong uptake of VCF and a successful transition of a large portion of its top customers to the new model. By 2Q25, over 87% of the 10,000 largest VMware customers had migrated from perpetual to subscription licence, driving double-digit ARR (annual recurring revenue) growth and locking in high renewal rates.
 - ⇒ **High margin:** Segment gross margin reached 93% in 2Q25 (up from 88% a year prior), with operating margin at 76% (up from 60% a year prior). The margin improvement reflects the disciplined integration of VMware. VMware's shift to subscription is the dominant EBIT and FCF lever, with management guiding for strong performance driven by ongoing transition to VCF subscriptions.
 - ⇒ **Portfolio rationalisation:** Broadcom has streamlined the VMware product portfolio, focusing on what it deems core offerings and divesting non-core assets. This is in line with its broader strategy of focusing on high-value, enterprise-critical software.

Transition from perpetual to subscription licencing

VMware officially ended sales of new perpetual (one-time purchase) licences and also stopped selling Support & Subscription (SnS) renewals for those legacy licences in Dec 2023. Existing perpetual keys remain valid but cannot be upgraded, combined or moved between versions. VMware's new subscription bundles (i.e. VMware Cloud Foundation and vSphere Foundation) are delivered entirely via subscription, offering optional add-ons (like storage or security). The move aligns VMware with the broader software industry's pivot to subscription models, aiming for predictable recurring revenue, continuous updates, and a simpler go-to-market approach. The change triggered backlash over cost increases and limited downgrade capability. Despite concerns, Broadcom reports over 87% of the 10,000 largest VMware customers had migrated from perpetual to subscription contracts.

Figure 35 - Software market size

Source: IDC, Macquarie Research, Sep 2025

Figure 36 - Infrastructure software revenue estimates

Source: Company data, Macquarie Research, Jul 2025

Software market sensitivity

The global software market may grow from ~US\$1.1tn in 2024 to ~US\$1.6tn by 2027, a 13% CAGR, according to IDC. Following the VMware acquisition, we estimate Broadcom's share of the software market to rise from ~0.8% in 2023 to around 2% in 2024, and continue to sustain the 2% market share by 2027.

To illustrate the potential outcomes, we present a sensitivity table of Broadcom's FY27E software revenue under varying assumptions for total software market size and Broadcom's market share. In our base case (~US\$1.6tn market, ~2% share), software rev would reach US\$30bn. In a downside case (~US\$1.3tn market, ~1% share), revenue would be US\$13bn, while in an upside case (~US\$1.9tn market, ~3% share), revenue could reach ~US\$57bn. We further extend this analysis to EPS impact. In our base case, FY27E EPS would reach US \$10.1. Under a downside case, EPS would be US\$7.6, while an upside case could lift EPS to US\$14.0.

We believe the company has the ability to operate its existing software assets more efficiently and potentially pursue further acquisitions in this segment, suggesting meaningful upside potential beyond our base assumptions. We view

Figure 37 - Broadcom software revenue sensitivity in FY27E

		FY27 TAM in US\$bn				
		1,300	1,500	1,600	1,700	1,900
AVGO	1.0%	13	15	16	17	19
	1.5%	20	23	24	26	29
	2.0%	26	30	32	34	38
	2.5%	33	38	40	43	48
	3.0%	39	45	48	51	57

Source: Company, IDC, Macquarie Research, Sep 2025

Figure 38 - Broadcom EPS sensitivity in FY27E

		FY27 TAM in US\$bn				
		1,300	1,500	1,600	1,700	1,900
AVGO	1.0%	9.8	10.1	10.2	10.4	10.6
	1.5%	10.7	11.1	11.4	11.6	12.0
	2.0%	11.7	12.3	12.6	12.8	13.4
	2.5%	12.6	13.3	13.7	14.0	14.7
	3.0%	13.5	14.4	14.8	15.3	16.1

Source: Company, IDC, Macquarie Research, Sep 2025

Strategic growth through M&A

With a proven track record of successful acquisitions, Broadcom has consistently turned opportunities into engines of expansion and profitability

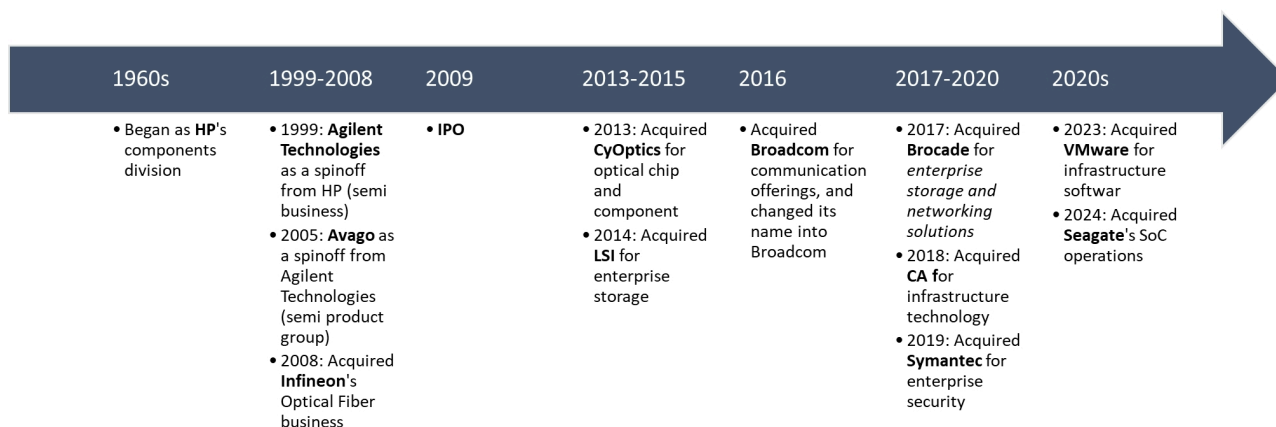
Since 2015, Broadcom has executed over US\$100 billion in acquisitions, with several deals reshaping the company's business mix and competitive position:

- **LSI (2014, US\$6.6bn):** Entry into enterprise storage and networking chips
- **Broadcom (2016, US\$37bn):** The transformative deal that prompted Avago to rebrand as Broadcom, establishing the company's scale in wired/wireless communications.
- **Brocade (2017 US\$5.5bn):** Fibre Channel networking which added enterprise storage connectivity
- **CA Technologies (2018, US\$18.9bn):** Marked Broadcom's important move into software, with stable mainframe-oriented cash flows.
- **Symantec Enterprise (2019 US\$10.7bn):** Added enterprise cybersecurity to the infrastructure software portfolio.
- **VMware (2023 US\$69bn):** The most ambitious deal to date, positioning Broadcom as a key player in private/hybrid cloud infrastructure and expanding its software segment to over 40% of total revenue.

Each of these acquisitions has followed the same integration philosophy: maintain the core product and customer value, centralise functions to cut operating expenses, and drive margin uplift through scale and shared infrastructure. Notably, Broadcom has consistently met or exceeded synergy targets and expanded operating margins post-acquisition.

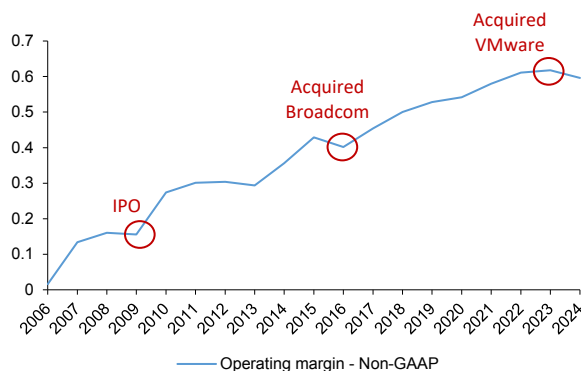
- **Long-term leadership alignment strengthens M&A execution.** In 2023, Broadcom granted CEO Hock Tan a US\$160.5mn performance-based stock award, structured to cover five years of future equity and cash compensation, with no eligibility for additional incentives until 2027. We view this front-loaded design as a deliberate move to reinforce executive continuity and ensure strategic focus across Broadcom's multi-year M&A roadmap. With Tan instrumental in past integrations (CA, Symantec Enterprise) and now leading VMware's transformation, the incentive structure aligns management rewards with shareholder value creation.

Figure 39 - Key M&A Deals



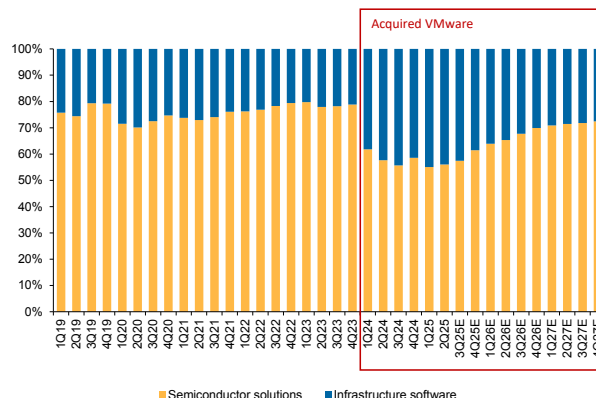
Source: Factset, Macquarie Research, Sep 2025

Figure 40 - Operating margin (Non-GAAP)



Source: Company data, Macquarie Research, Sep 2025

Figure 41 - Revenue Segment (USD'mn)

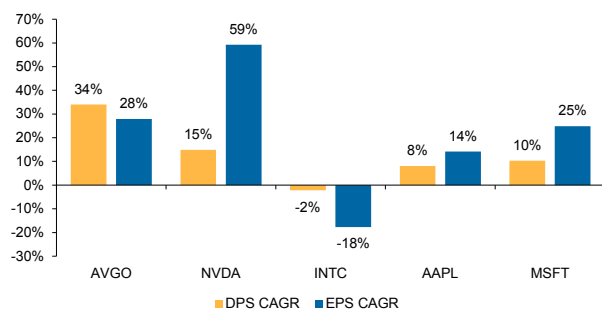


Source: Company data, Macquarie Research, Sep 2025

Capital allocation analysis

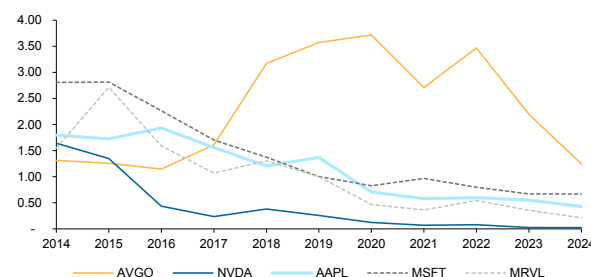
Broadcom has transformed into a shareholder cash-returns machine under CEO Hock Tan. The company initiated its first dividend in 2011 and has increased it every year since, achieving a ~34% dividend CAGR over the past decade (2014-2024), outpacing earnings growth over the same period (EPS CAGR: 28%), pushing the payout ratio from 21% in 2011 to about 64% in 2024. Management has repeatedly affirmed its commitment to a sizeable dividend. Broadcom targets returning ~50% of prior year free cash flow to shareholders in the form of cash dividends. The remaining free cash flow is allocated to a combination of share repurchases, debt reduction, and opportunistic M&A, with a strong bias toward maintaining balance sheet flexibility.

Figure 42 - DPS and EPS CAGR in 2014-2024



Source: Factset, Macquarie Research, Sep 2025

Figure 43 - Dividend yield against peers



Source: Factset, Macquarie Research, Sep 2025

Supply chain

Key supply resides in Taiwan

Broadcom relies heavily on Taiwanese firms for chip fabrication and packaging.

- **TSMC** is Broadcom's dominant foundry partner, fabricating over 90% of Broadcom's wafers. TSMC manufactures Broadcom's advanced networking ASICs, custom silicon, and wireless SoCs on leading nodes.
- **ASE Technology / SPIL**: Among Broadcom's principal OSATs, handling a wide range of packages (flip-chip, WLCSP, SiP) and final test.
- **Foxconn**: Supports module assembly, board-level build, and chip packaging.

- **King Yuan Electronics (KVEC):** Taiwan's largest pure-play testing house. Provides wafer probing, final test, and burn-in for logic, RF/wireless, and mixed-signal chips. Major customers include Broadcom and NVIDIA.
- **PCL Technologies (4977 TT, Not Rated):** Optical transceiver and CPO vendor. PCL recently won Broadcom's laser-optical packaging orders for its latest data-center switch ASIC CPO modules (see news [here](#)).
- **MPI Corporation (6223 TT, Not Rated):** Leading Taiwan probe card and probe station supplier. Its advanced RF/mmWave and photonics test solutions are deployed at Broadcom's OSAT partners (e.g., ASE, KVEC), enabling wafer-level validation and optical/RF device testing.
- **Unimicron:** Major ABF substrate supplier. Provides IC substrates (via OSAT) for Broadcom's high-pin-count ASICs and networking chips.

Figure 44 - Key Taiwan suppliers

Company	Relationship	Supplied Good/Service	Product Line
TSMC	Direct	Wafer fabrication; ~95% of Broadcom wafers	Advanced ASICs, SoCs
ASE / SPIL	Direct	Chip assembly & test	Semiconductors & Hardware
Foxconn	Direct	Assembly/test; module & board build	Semiconductors & Hardware
KVEC	Direct	Wafer probe, final test, burn-in	RF, SoC, mixed-signal
PCL Technology	Direct	Laser-optical packaging for CPO	Data center switching optics
Unimicron	Sub-tier	ABF IC substrates	Advanced packaging
MPI	Sub-tier	Probe cards, probe stations, photonics/RF test equipment	Wafer/device test enablement

Source: Macquarie Research, Sep 2025



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Memory mainly sourced from South Korea

South Korean firms primarily supply Broadcom with memory and advanced components rather than core logic chips.

- **Samsung Electronics:** Broadcom sourced HBM3 exclusively from Samsung, while HBM3E is supplied by both Samsung and SK Hynix. In addition, Samsung is partnering with Broadcom on next-generation silicon photonics technology, a forward-looking collaboration aimed at integrating optical I/O into Broadcom's future ASICs, suggesting Samsung could become a foundry/technology partner for Broadcom's optical networking chips within the next two years (see news [here](#)).
- **SK Hynix:** A major supplier of HBM for Broadcom's AI accelerators. In late 2024, Broadcom placed a large order with SK Hynix to procure HBM memory chips for ASICs (see news [here](#)).
- **Samsung Electro-Mechanics (SEMCO):** Supplier of advanced ABF substrates and MLCC capacitors.

Figure 45 - Key South Korea suppliers

Company	Relationship	Supplied Good/Service	Product Line
Samsung Electronics	Direct supplier	HBM chips integrated on ASIC packages	Semiconductors (AI/Networking)
Samsung Electronics	Future supplier	(Planned) Foundry & R&D partner for silicon photonics; potential fab for optical ASICs	Semiconductors (optical ASICs)
SK Hynix	Direct supplier	HBM chips integrated on ASIC packages	Semiconductors (AI/Networking)
Samsung Electro-Mechanics	Sub-tier	IC substrates and passive components (MLCC capacitors) for packaged devices (via OSATs or EMS)	Semiconductors; Hardware

Source: Macquarie Research, Sep 2025

China provides assembly base and rare earth materials

Broadcom's direct supplier base in China is relatively limited. However, China is a significant location for manufacturing assembly of Broadcom's products via partners. Notably, Broadcom utilises Foxconn's extensive China-based facilities for assembling both



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semiconductor packages and finished hardware. Additionally, Broadcom's manufacturing relies on a variety of raw materials (including rare earth elements, copper, and speciality metals), and China dominates the supply of many of these inputs. Any disruption in China's materials exports (e.g. export controls on rare earths) could therefore impact Broadcom. Trade tensions or supply constraints in China may lead to extended lead times and shortages in materials like substrates and wafers.

Japan for materials and substrates

Japan's contribution is materials (substrates, optics, RF) and emerging foundry diversification via Rapidus.

- **Rapidus (unlisted)**: In early 2025, reports (see news [here](#)) surfaced that Rapidus would provide Broadcom with 2-nanometer prototype wafer runs starting in June 2025. This is part of Japan's effort to establish an advanced foundry alternative to TSMC. At Hot Chip 2025, Rapidus announced the successful 2nm GAA test chip tape-out and plans to begin mass production by 2027.
- **Toppan (7911 JP, Not Rated)**: Leading ABF substrate vendor for high-end ASIC packages. Broadcom's chiplet designs rely on Toppan substrates via Broadcom's OSAT partners. Currently, FC-BGA for Broadcom's network switches is being shipped from the Niigata factory. Furthermore, Broadcom and Toppan are investing together in a factory in Singapore (an FC-BGA factory), which is scheduled to begin mass production in early 2027. According to media [reports](#), the capital investment is ¥50bn, with Broadcom and Toppan each contributing 50%.
- **Murata mfg** supplies RF modules for smartphones and competes with Broadcom. However, Murata also supplies RF devices, such as filters and duplexers, to Broadcom. In this case, the relationship is that of both customer and supplier.
- Companies related to optical transceivers: Japanese optical semiconductor makers (both laser and photodiode) are Broadcom's competitors and potential suppliers. **Sumitomo Electric** and **Furukawa Electric** (5801 JP, Not Rated) have the opportunity to supply Broadcom with long-range lasers (LR4, ER4, and ZR). **Dexerials'** (4980 JP, Not Rated) production capacity for photodiodes is small, but Broadcom may use Dexerials' technology for co-packaged optics (CPO). **Maruwa** (5344 JP, Not Rated) has been supplying a ceramic substrate for heat dissipation for optical modules.

Figure 46 - Key Japan suppliers

Company	Relationship	Supplied Good/Service	Product Line
Rapidus	Future direct	Advanced foundry for 2nm	Semiconductors
Toppan	Direct / Sub-tier	High-end ABF package substrates (via OSATs)	Semiconductors
Sumitomo Electric	Sub-tier	Optical laser	Optical
Furukawa Electric	Sub-tier	Optical laser	Optical
Dexerials	Sub-tier	Optical photo diode	Optical
Maruwa	Sub-tier	Ceramic substrate for optical devices	Optical
Murata Mfg	Direct / Sub-tier	RF module / RF device	RF

Source: Macquarie Research, Sep 2025



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ASEAN / Malaysia for OSAT

Inari is a semiconductor backend outsourced assembly and testing (OSAT) company, the largest in Malaysia by revenue. Inari has factories in Malaysia, China and the Philippines. Inari's largest customer is Broadcom. It has been trying to diversify away from Broadcom evident from its recent acquisition of Lumineds. Its exposure to Broadcom is ~90% of its revenue which are:

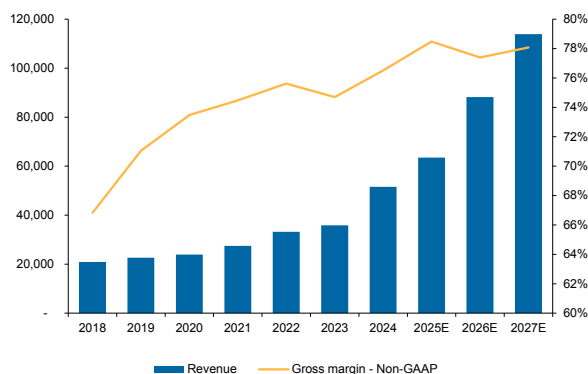
- **RF**: ~60% of sales, primarily for assembly and testing of RF filters used in smartphones (Android and iPhones). Inari's RF revenue has >90% correlation to Broadcom's wireless revenue.
- **Opto**: ~20% of sales, primarily for testing optoelectronics components which includes fibre optic transceivers from its Malaysia and Philippines facilities.

Financial analysis

Broadcom delivered record top-line and margin expansion since FY24, with AI semis and VMware driving a step-change in scale and profitability.

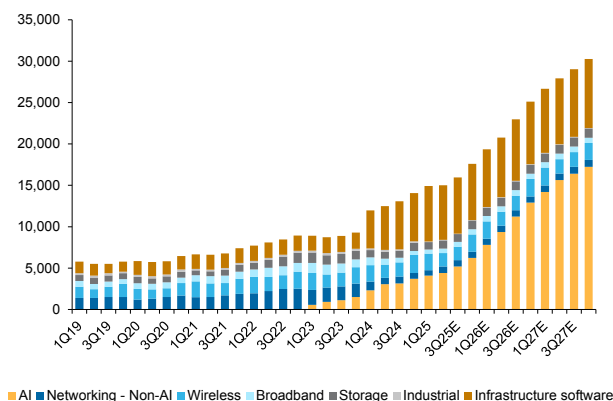
- **Revenue** base is now structurally more diversified, with software and AI semis driving both scale and resilience, while non-AI semis were flat to down YoY. FY24 revenue surged 44% YoY to US\$51.6bn, with 2Q25 at US\$15.0bn (20% YoY).
 - ⇒ Semiconductor Solutions posted US\$30.1bn FY24 revenue (7% YoY), with AI semis exceeding US\$12bn. 2Q25 AI semis hit US\$4.4bn (44% YoY). Semiconductor Solutions gross margin reached 69% in 2Q25 (up 140 bps YoY).
 - ⇒ Infrastructure Software revenue jumped to US\$21.5bn in FY24 (up 181% YoY, or 19% ex-VMware). 2Q25 software gross margin reached 93% (up 500 bps YoY).
- **Margin** expansion is structurally underpinned by the pivot to high-margin software, AI semi content, and tight cost control, with further upside as VMware synergies mature
 - ⇒ Gross margin (Non-GAAP) expanded to 76.5% in FY24 (vs 74.7% FY23) as infrastructure software (93% gross margin) grew to 42% of revenue post-VMware acquisition. 2Q25 gross margin was 79.4%.
 - ⇒ Operating margin (Non-GAAP) climbed from 61% in FY22 to 65% in 2Q25, reflecting both software leverage and AI semi content gains. Despite R&D spend (non-GAAP) increased by 56% to US\$5.8bn in FY24 post-VMware acquisition, opex/revenue ratio continued to decline from 18.3% in 1Q24 to 14.1% in 2Q25 as VMware integration delivered margin accretion ahead of schedule.

Figure 47 - Revenue and Gross Margin (USD'mn)

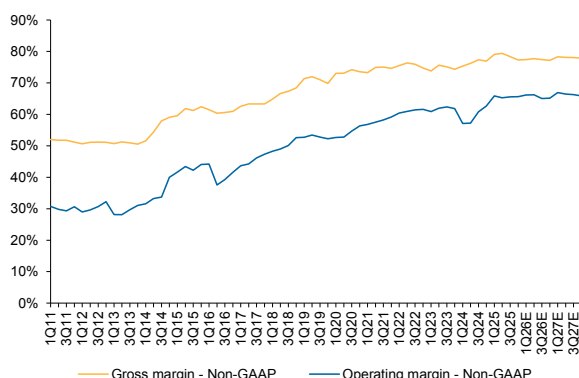


Source: Company data, Macquarie Research, Sep 2025

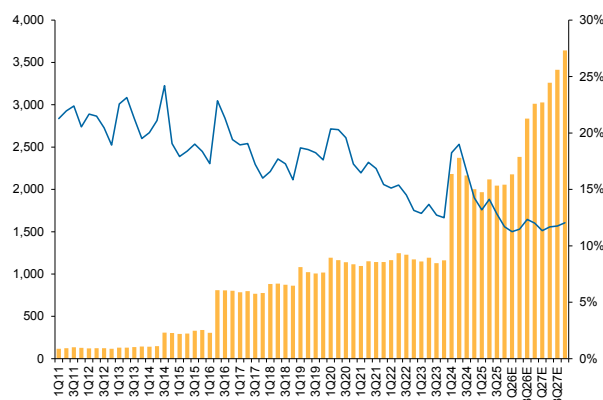
Figure 48 - Revenue by segment (USD'mn)



Source: Company data, Macquarie Research, Sep 2025

Figure 49 - Gross margin and operating profit margin (non-GAAP)

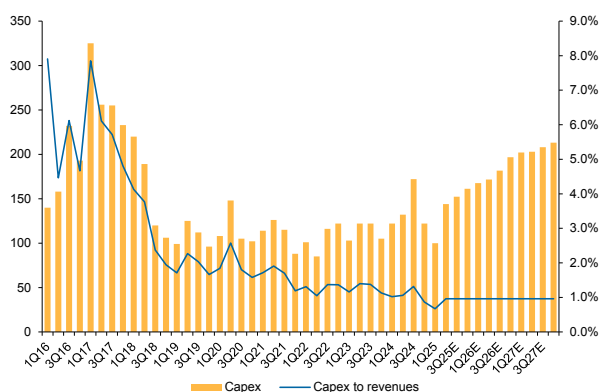
Source: Company data, Macquarie Research, Sep 2025

Figure 50 - Operating expense and as % of revenue (USD'mn)

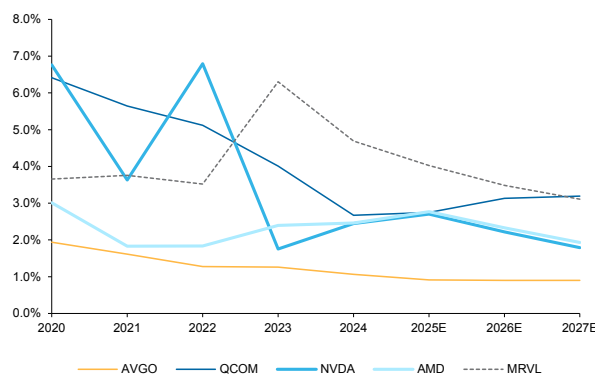
Source: Company data, Macquarie Research, Sep 2025

Capex remains minimal at US\$548mn in FY24 (~1% of revenue), consistently below depreciation (US\$593mn), reflecting Broadcom's fabless, capital-light model. Capex rose modestly in FY24 due to VMware integration and incremental IT infrastructure, but remains immaterial to cash flow or margin structure.

- Over the past 5 years, capex as a percentage of revenue averaged 1.0%–2.0%, compared to 3–6% for Qualcomm and 2%–6% for Nvidia, highlighting Broadcom's superior capital efficiency among peers.
- Node migration and capacity reservations are managed through foundry agreements and long-term supply contracts, with no need for direct fab investment.
- We expect capex requirements to remain at about 1.0% of revenue, even as AI and custom silicon volumes scale, given the continued reliance on external foundries and minimal physical plant expansion.

Figure 51 - Capex and capex intensity (USD'mn)

Source: Company data, Macquarie Research, Jul 2025

Figure 52 - Capex intensity vs peers

Source: Company data, Factset, Macquarie Research, Sep 2025

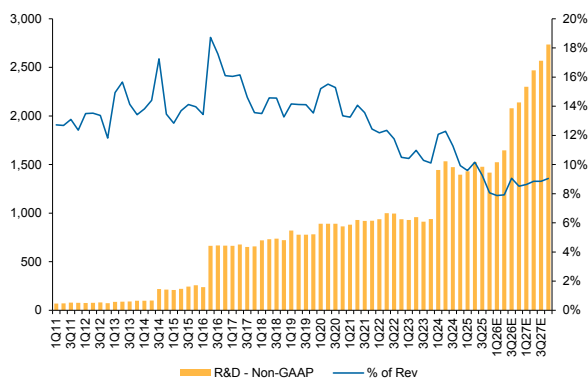
R&D engine is growing rapidly with spend, headcount, and patent output all surging to support AI and custom silicon leadership.

- **R&D expense** increased by 56% to US\$5.8bn (11% of revenue) in FY24, up from US\$3.7bn (10% of revenue) in FY23, reflecting VMware integration and AI investment acceleration. Quarterly R&D run-rate now exceeds US\$1.5bn, with 2Q25 at US\$1.5bn. R&D CAGR (FY21–24) is 17%. R&D intensity now exceeds NVDA. Patent portfolio expanded sharply post-VMware. Total patents rose to 20,870 in FY24 after acquisition of VMware, with 2,650 pending applications.

- **R&D headcount** pre-VMware was over 12,900 (64% of 20,000 total employees in FY23). Post-VMware, total workforce exceeds 37,000, with over 20,000 R&D staff. The VMware integration added significant software engineering talent, further diversifying Broadcom's technical base and deepening its presence in enterprise infrastructure. R&D focus is shifting toward AI, custom silicon, and cloud infrastructure, with legacy connectivity and mainframe software maintained for cash flow. It has a dedicated team for AI/custom silicon project, with over 1,000 engineers just dedicated to SerDes design. Broadcom is determined to out-invest the industry in core compute, networking, and memory IP.
- **Output** is visible in pipeline wins. Broadcom doubled switch chip bandwidth every two years, secured custom AI accelerator deals with hyperscalers, and leads in high-speed networking for AI clusters. Revenue per staff trends upwards from 2015 onwards. Management expects further efficiency gains as VMware integration synergies scale and AI-driven product cycles mature, supporting both margin expansion and IP growth

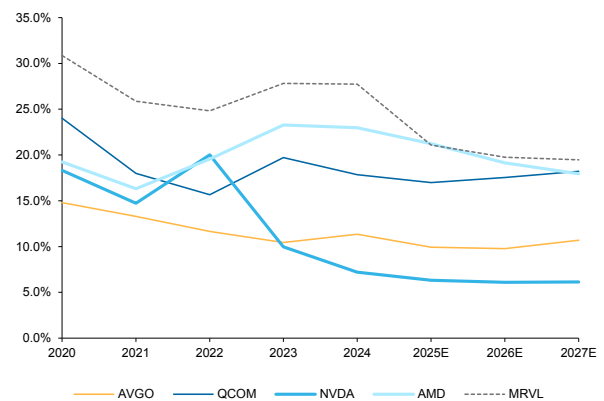
Broadcom's R&D platform is now scaled for AI and software leadership, with patent firepower, engineering depth, and capital intensity all supporting durable competitive advantage

Figure 53 - R&D and R&D intensity (USD'mn)



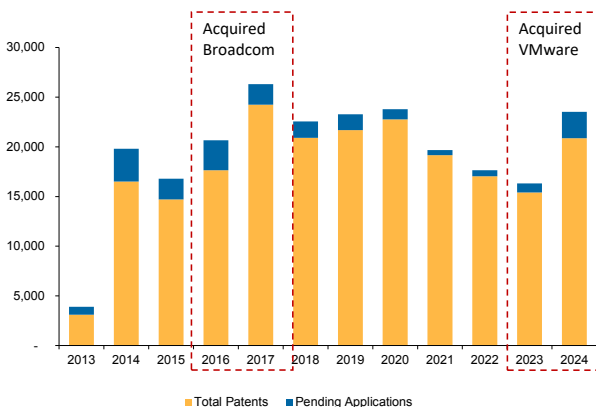
Source: Company data, Macquarie Research, Sep 2025

Figure 54 - R&D intensity vs peers



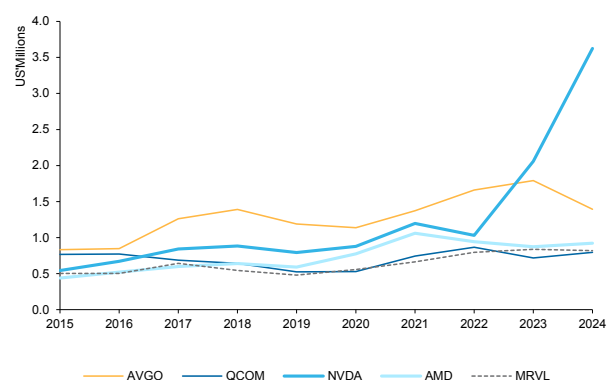
Source: Company data, Factset, Macquarie Research, Sep 2025

Figure 55 - Broadcom's patents



Source: Company data, Factset, Macquarie Research, Sep 2025

Figure 56 - Output efficiency (revenue per staff)



Source: Company data, Factset, Macquarie Research, Sep 2025

Valuation

Figure 57 - Weighted average P/E

2027E	OP mix	PEx
Semiconductor solutions	65%	23.3
Infrastructure software	35%	52.5
Weighted avg		33.5

Source: Company data, Factset, Macquarie Research, Sep 2025

We value Broadcom using a sum-of-the-parts (SOTP) approach that reflects the structurally different profiles of its two core business: semiconductor solutions and infrastructure software. While software currently contributes 40% of revenue, we estimate it contributes roughly 35% of operating profit in 2027E due to its higher margins and capital efficiency. These warrant differentiated valuation multiples. We benchmarked against the top US listed companies by market cap in each respective sector:

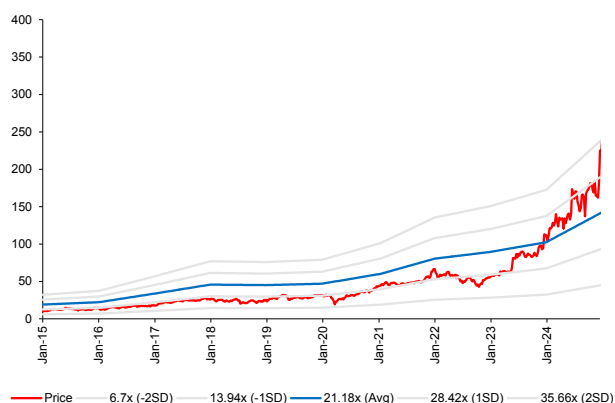
- Semiconductor peer average: 2027E forward P/E of 23.3x.
- Software peer average: 2027E forward P/E of 52.5x

Applying these to Broadcom's FY27E EPS (65% weight for semi and 35% weight for software), we derive a blended implied multiple of 33.5x. Our FY27E EPS estimate is US \$12.55. We derive a target price of US\$420, implying a 17% TSR from the current level.

This valuation reflects:

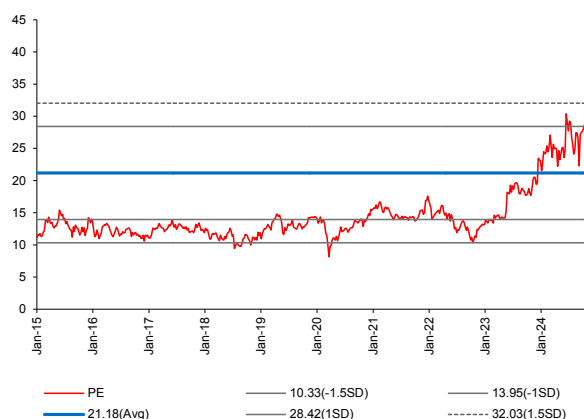
- The defensible moat and pricing power in Broadcom's semi portfolio (custom ASICs, switching, etc.)
- The margin accretion, recurring revenue, and multiple expansion potential from VMware and broader software assets.

Figure 58 - 12-month forward PE bands (average since 2022)



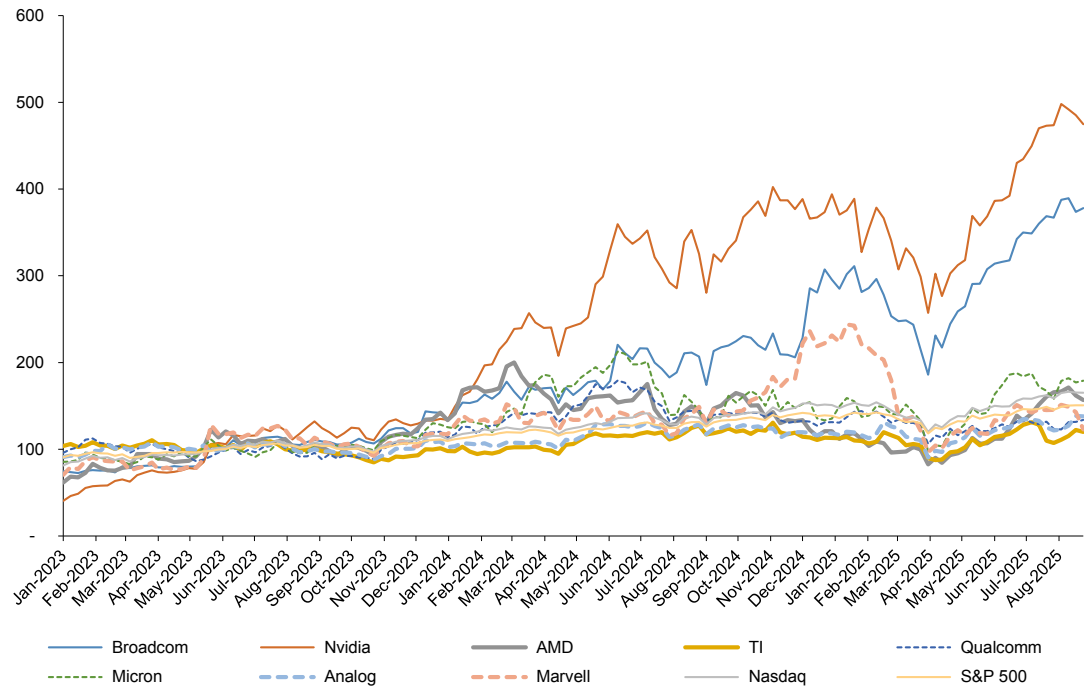
Source: Company data, Macquarie Research, Sep 2025

Figure 59 - 12-month forward PE bands (average since 2022)



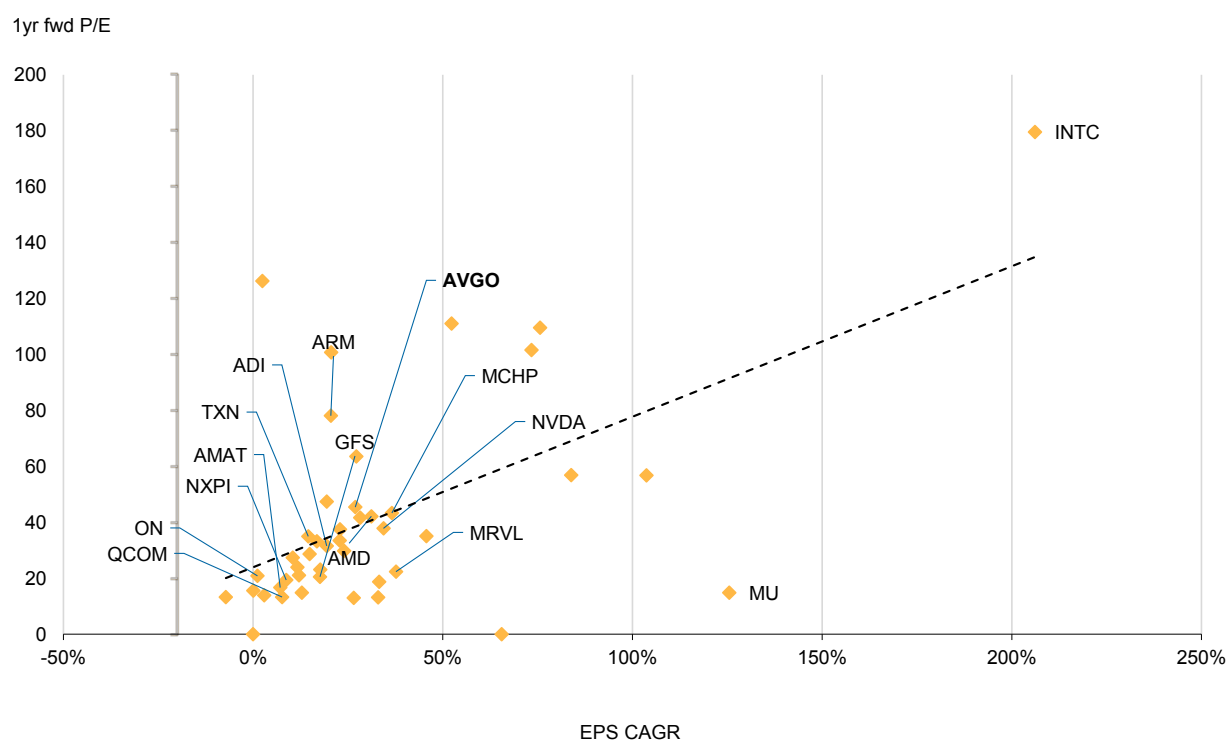
Source: Company data, Macquarie Research, Sep 2025

Figure 60 - Broadcom stock price vs chip design peers (indexed to 2023 average)

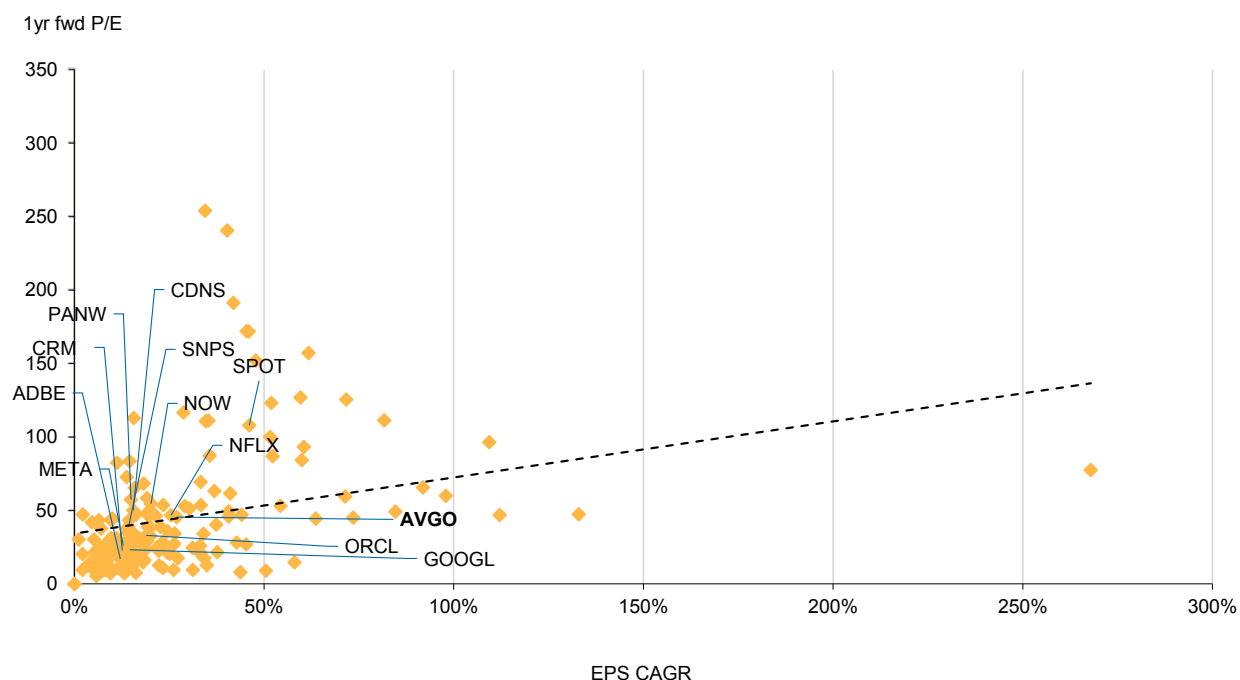


Source: Factset, Macquarie Research, Sep 2025

Figure 61 - US Semiconductor Stocks EPS CAGR vs. Forward P/E Multiples Chart



Source: Factset, Macquarie Research, Sep 2025

Figure 62 - US Software Stocks EPS CAGR vs. Forward P/E Multiples Chart

Source: Factset, Macquarie Research, Sep 2025

Figure 63 - Semiconductor valuation comparisons

Ticker	Company	Market cap US\$m	Price US\$	EPS (US\$)			P/E			EBITDA (US\$m)			EV/EBITDA		
				2025E	2026E	2027E	2025E	2026E	2027E	2025E	2026E	2027E	2025E	2026E	2027E
AVGO	Broadcom Inc.	1,698,304	359.63	6.74	9.17	11.72	53.33	39.22	30.69	42,110	57,560	69,270	40.33	29.51	24.52
NVDA	NVIDIA Corporation	4,305,231	177.17	4.52	6.37	7.33	39.23	27.81	24.18	129,612	185,279	215,239	33.22	23.24	20.00
AMD	Advanced Micro Devices, Inc.	252,628	155.67	3.85	5.95	7.46	40.45	26.17	20.87	6,627	10,120	12,367	38.12	24.96	20.43
TXN	Texas Instruments Incorporated	167,599	184.35	5.60	6.67	7.81	32.93	27.65	23.59	8,146	9,667	10,986	20.58	17.34	15.26
QCOM	QUALCOMM Incorporated	174,269	161.51	11.83	12.21	12.97	13.65	13.23	12.46	16,981	16,319	16,425	10.26	10.68	10.61
AMAT	Applied Materials, Inc.	135,549	170.15	9.38	9.52	10.63	18.14	17.88	16.00	8,994	9,175	10,105	15.07	14.77	13.41
ARM	ARM Holdings PLC Sponsored ADR	163,827	154.70	1.68	2.25	2.85	91.93	68.84	54.26	2,252	2,983	3,892	72.76	54.92	42.10
MU	Micron Technology, Inc.	168,507	150.57	8.02	13.02	14.90	18.78	11.57	10.11	18,438	26,884	29,443	9.14	6.27	5.72
ADI	Analog Devices, Inc.	122,123	248.24	7.75	9.36	10.86	32.03	26.53	22.86	4,927	5,717	6,427	24.79	21.36	19.00
INTC	Intel Corporation	114,935	24.61	0.13	0.69	1.25	187.46	35.70	19.64	12,393	16,662	21,232	9.27	6.90	5.41
MRVL	Marvell Technology, Inc.	57,407	66.59	2.79	3.35	4.10	23.85	19.88	16.26	3,034	3,488	3,949	18.92	16.46	14.54
NXPI	NXP Semiconductors NV	56,275	223.21	11.78	14.00	16.82	18.94	15.94	13.27	4,554	5,204	5,922	12.36	10.81	9.50
TEL	TE Connectivity plc	62,760	212.40	8.60	9.51	10.54	24.71	22.34	20.16	4,177	4,666	4,978	15.03	13.45	12.61
MCHP	Microchip Technology Incorporated	35,090	65.02	1.46	2.55	3.34	44.52	25.52	19.48	1,873	2,279	2,970	18.74	15.40	11.81
MPWR	Monolithic Power Systems, Inc.	40,247	840.38	17.46	20.23	24.03	48.14	41.54	34.97	957	1,117	1,315	42.04	36.03	30.61
ON	ON Semiconductor Corporation	20,048	49.02	2.29	2.93	4.12	21.37	16.76	11.90	1,600	1,871	2,376	12.53	10.71	8.44
GFS	GlobalFoundries Inc.	17,787	32.17	1.62	1.99	2.54	19.88	16.15	12.67	2,316	2,577	2,994	7.68	6.90	5.94
FSLR	First Solar, Inc.	21,805	203.31	15.33	23.25	28.24	13.26	8.74	7.20	2,270	3,256	3,823	9.61	6.70	5.70
ALAB	Astera Labs, Inc.	38,712	232.90	1.59	2.05	2.97	146.44	113.49	78.49	256	347	532	151.51	111.42	72.74
SWKS	Skyworks Solutions, Inc.	11,085	74.68	5.57	4.25	5.02	13.41	17.58	14.89	1,280	929	1,157	8.66	11.93	9.58
FN	Fabrinet	12,940	362.16	10.17	12.18	14.17	35.61	29.74	25.56	414	497	593	31.25	26.01	21.82

Source: Factset, Macquarie Research, Sep 2025. Priced on 11 Sep 2025.

Figure 64 - Software valuation comparisons

Ticker	Company	Market cap US\$m	Price US\$	EPS (US\$)			P/E			EBITDA (US\$m)			EV/EBITDA		
				2025E			2025E			2025E			2025E		
NFLX	Netflix, Inc.	511,399	1,203.50	26.28	32.29	39.27	45.80	37.27	30.65	14,146	17,127	20,222	36.15	29.86	25.29
PLTR	Palantir Technologies Inc. Class A	373,798	164.36	0.64	0.84	1.13	255.09	195.93	145.25	1,941	2,685	3,726	192.56	139.20	100.32
NOW	ServiceNow, Inc.	195,162	938.28	16.89	20.03	24.17	55.56	46.84	38.83	4,594	5,566	6,854	42.49	35.06	28.47
SPOT	Spotify Technology SA	142,050	690.31	6.49	13.35	17.81	106.45	51.70	38.75	2,583	3,576	4,736	55.00	39.72	29.99
PANW	Palo Alto Networks, Inc.	132,663	198.33	3.34	3.81	4.31	59.38	52.12	46.06	2,890	3,341	3,766	45.90	39.71	35.22
CRWD	CrowdStrike Holdings, Inc. Class A	108,759	433.38	3.66	4.75	6.08	118.26	91.23	71.29	1,268	1,677	2,094	85.78	64.84	51.93
CDNS	Cadence Design Systems, Inc.	96,652	354.70	6.93	7.96	9.20	51.15	44.57	38.56	2,460	2,808	3,215	39.28	34.43	30.06
SNPS	Synopsys, Inc.	81,376	438.10	12.95	14.20	17.44	33.84	30.86	25.12	2,786	4,277	4,947	29.21	19.02	16.45
SNOW	Snowflake, Inc.	76,071	224.53	1.20	1.65	2.38	187.07	136.12	94.44	484	643	901	157.01	118.36	84.41
CRWV	CoreWeave, Inc. Class A	41,748	112.69	(1.62)	(0.44)	2.02	N/A	N/A	55.85	3,396	8,529	12,686	12.29	4.90	3.29
TEAM	Atlassian Corp Class A	29,384	176.71	3.68	4.26	5.22	48.02	41.49	33.87	1,316	1,572	2,001	22.32	18.69	14.69
DDOG	Datadog, Inc. Class A	44,983	139.15	1.83	2.15	2.68	76.21	64.79	51.86	762	962	1,238	59.00	46.77	36.32
ZS	Zscaler, Inc.	44,632	286.66	3.28	3.65	4.39	87.40	78.50	65.23	682	863	1,068	65.47	51.71	41.80
CRCL	Circle Internet Group, Inc. Class A	28,323	133.70	(1.70)	1.28	2.86	N/A	104.10	46.72	480	592	1,020	58.96	47.86	27.76
TTD	Trade Desk, Inc. Class A	20,162	45.24	0.82	1.08	1.37	55.08	41.86	32.91	1,122	1,338	1,576	17.98	15.07	12.79
CSGP	CoStar Group, Inc.	36,883	87.06	0.79	1.35	1.96	109.69	64.33	44.39	389	734	1,084	94.78	50.27	34.01
FWONK	Liberty Media Corp. Series C Liberty F	34,591	102.14	2.31	2.08	2.62	44.22	49.06	39.04	1,001	1,251	1,333	34.55	27.66	25.94
HUBS	HubSpot, Inc.	26,177	496.79	9.53	11.43	14.00	52.14	43.45	35.49	687	836	1,057	38.09	31.29	24.77
RDDT	Reddit, Inc. Class A	35,065	260.54	1.84	3.17	4.23	141.87	82.30	61.63	734	1,093	1,486	47.77	32.09	23.59
CYBR	CyberArk Software Ltd.	23,751	471.19	3.90	4.88	6.46	120.70	96.47	72.91	313	393	426	75.82	60.41	55.79
MDB	MongoDB, Inc. Class A	27,019	332.10	3.75	4.36	5.50	88.58	76.16	60.35	344	429	615	78.44	62.98	43.91

Source: Factset, Macquarie Research, Sep 2025. Priced on 11 Sep 2025.

Key risks

- **AI cycle risk:** While current order books and tech roadmap point to continued market share dominance through at least 2027, hyperscaler AI capex normalisation could slow growth.
- **VMware's integration execution.** Delays in synergy capture, cost-out, or cross-sell could compress software margin and FCF, slowing deleveraging and risking negative estimate revisions.
- **Supply chain risk.** Export controls, supply chain shocks, or China demand weakness could disrupt both semi and software revenue.
- **Regulatory.** Ongoing scrutiny of VMware pricing, software bundling, and cross-border data flows could trigger fines, forced divestitures, or margin headwinds.
- **Client concentration.** Top hyperscalers contribute the majority of AI semi revenue. Loss of a major design win or insourcing risk could materially impact growth and margin.

Key Quant Findings (top 5% of the Macquarie Quant Alpha model)

The quant model currently holds a strong positive view on Broadcom. The strongest style exposure is Price Momentum, indicating this stock has had strong medium to long term returns which often persist into the future. The weakest style exposure is Profitability, indicating this stock is not efficiently converting investments to earnings; proxied by ratios like ROE or ROA.

Macquarie Alpha Model: Key rankings

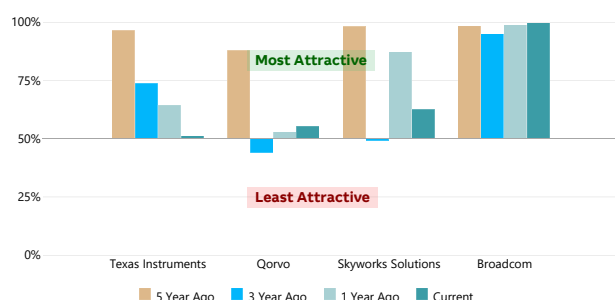
The Macquarie Quant's flagship Alpha model is a dynamic multi-factor model based on a staple of quant factors such as value, momentum, revisions, quality, and risk.

	Global	Market (Country)	Sector
	Whole Universe	United States	Semiconductors & Semiconductor Equip.
Macquarie Alpha Model	39/17713	9/3089	2/664
Fundamental (Consensus) *	2993/17713	409/3089	135/664

* based on Total Shareholder Return = Consensus Price target / Current Price

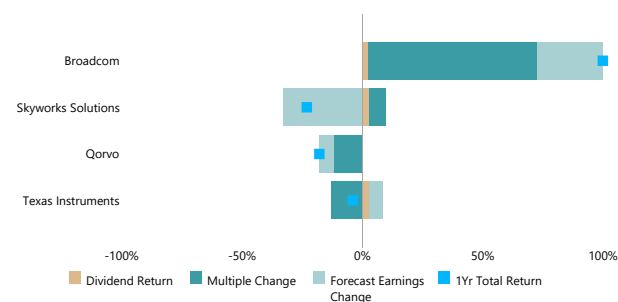
Current and Historical Alpha Model Rank

The chart shows the Macquarie Alpha model market ranking against the company's peers and over recent history.



Drivers of Stock Return

Breakdown of 1-year total return (local currency) into returns from dividends, changes in forward earnings estimates and the resulting change in earnings multiple.



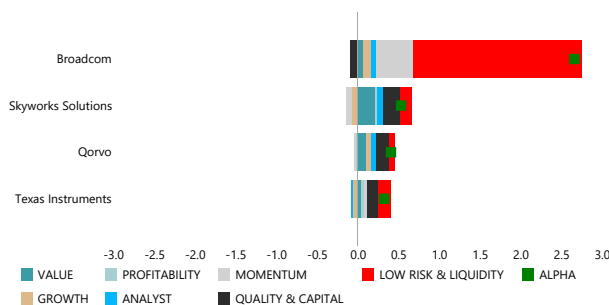
Alpha Model Decomposition

The Macquarie Alpha is decomposed into its sector and market relative factor & styles exposures (a higher/better percentile is coded in green, whilst lower in red).

Factors / Styles	Percentile relative to		Core factors in definition
	sectors (/664)	market (/3089)	
ALPHA	100%	100%	Built from the styles below
VALUE	69%	38%	Book, CF, Yield, Earnings Multiples
ANALYST	91%	84%	Revisions (Earnings, Recommendations)
MOMENTUM	98%	100%	Price Momentum
GROWTH	75%	82%	EPS, Sales (Forecast, Historic)
PROFITABILITY	41%	53%	ROE, Margin, Asset Turnover
QUALITY	16%	27%	Accruals, Earn Stability, Cash Conversion
CAPITAL	28%	28%	Investment/Capex, Net share issuance
LIQUIDITY	100%	100%	Size, Turnover, Analyst Coverage
LOW RISK	18%	22%	Beta, Volatility, Earn.Cert, Leverage
TECHNICAL	6%	4%	MACD, RSI, Bollinger, Williams R, etc

Factors driving the Alpha Model vs peers

For the comparable firms this chart shows the key underlying styles and their contribution to the current overall raw Alpha score.



Macquarie Style Returns over last year

Recent performance to Macquarie style factors

United States	Monthly Factor Long-Short Returns for												Last 5 Years (ann)	Last 10 Years (ann)
	Aug - 25	Jul - 25	Jun - 25	May - 25	Apr - 25	Mar - 25	Feb - 25	Jan - 25	Dec - 24	Nov - 24	Oct - 24	Sep - 24		
ALPHA	0%	-2%	1%	4%	4%	-3%	-3%	3%	2%	4%	2%	0%	9%	6%
VALUE	4%	-2%	0%	0%	-6%	5%	1%	1%	-1%	1%	2%	-2%	12%	3%
ANALYST	-1%	-5%	1%	5%	7%	-5%	-2%	4%	2%	2%	2%	1%	5%	3%
MOMENTUM	-6%	2%	0%	1%	3%	-1%	-2%	3%	-2%	7%	3%	0%	0%	1%
GROWTH	-2%	3%	4%	8%	7%	-5%	-3%	2%	3%	6%	2%	2%	9%	5%
PROFITABILITY	1%	-2%	-1%	1%	3%	-3%	-2%	1%	2%	2%	1%	-1%	2%	5%
QUALITY	1%	0%	-1%	3%	2%	-2%	-3%	-1%	2%	0%	-2%	-1%	0%	2%
CAPITAL	1%	-3%	0%	0%	-1%	0%	-1%	-1%	1%	-2%	-1%	-2%	4%	3%
LIQUIDITY	-1%	0%	3%	2%	1%	-3%	-1%	2%	4%	-1%	1%	-2%	0%	5%
LOW RISK	0%	-2%	-5%	0%	2%	-1%	-1%	1%	1%	1%	0%	-1%	-1%	2%

Source (all charts): FactSet, Refinitiv, and Macquarie Quant. For more details on the Macquarie Alpha model or for more customised analysis and screens, please contact the Macquarie Global Quantitative Team: macquarieequitiesresearchquantglobal@macquarie.com. Explanation for items on this page can be found at <https://www.macquarieinsights.com/rp/d/r/p/OTUyMzg1>

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Important Disclosures

Recommendation definitions	Volatility index definition	Financial definitions
Macquarie – Asia and USA Outperform – expected return >10% Neutral – expected return from -10% to +10% Underperform – expected return <-10% Macquarie – Australia/New Zealand Outperform – expected return >10% Neutral – expected return from 0% to 10% Underperform – expected return <0% During periods of share price volatility, recommendations and target prices may occasionally and temporarily be inconsistent with the above definitions. Recommendations – 12 months 12-month target – Expected share price in 12 months Valuation – The company's estimated fair value share price based on the disclosed valuation methodology Note: Quant recommendations may differ from Fundamental Analyst recommendations	This is calculated from the volatility of historical price movements. Very high – highest risk – Stock should be expected to move up or down 60–100% in a year – investors should be aware this stock is highly speculative. High – stock should be expected to move up or down at least 40–60% in a year – investors should be aware this stock could be speculative. Medium – stock should be expected to move up or down at least 25–40% in a year. Low – stock should be expected to move up or down at least 15–25% in a year. * Applicable to select stocks in Asia/Australia/NZ Note: expected return is reflective of a Medium Volatility stock and should be assumed to adjust proportionately with volatility risk	All "Adjusted" data items have had the following adjustments made: Added back: goodwill amortisation, provision for catastrophe reserves, IFRS derivatives & hedging, IFRS impairments & IFRS interest expense Excluded: non recurring items, asset revals, property revals, appraisal value uplift, preference dividends & minority interests EPS = adjusted net profit / <i>efpowa</i> [*] ROA = adjusted ebit / average total assets ROA Banks/Insurance = adjusted net profit / average total assets ROE = adjusted net profit / average shareholders funds Gross cashflow = adjusted net profit + depreciation [*] equivalent fully paid ordinary weighted average number of shares All Reported numbers for Australian/NZ listed stocks are modelled under IFRS (International Financial Reporting Standards).

Recommendation proportions for quarter ending 30 Jun 2025

	AU/NZ	Asia	USA	
Outperform	63.35%	65.78%	70.73%	(for global coverage by Macquarie, 1.03% of stocks followed are investment banking clients)
Neutral	30.60%	21.00%	29.27%	(for global coverage by Macquarie, 1.77% of stocks followed are investment banking clients)
Underperform	6.05%	13.23%	0.00%	(for global coverage by Macquarie, 0.00% of stocks followed are investment banking clients)

Company-Specific Disclosures

Company Name	Disclosure
Broadcom (AVGO US) Outperform 12-month target: USD420.00 - PER Valuation: USD 420.00 - PER Price: USD359.87	None

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15 September 2025

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